

Received 29 October 2024, accepted 6 December 2024, date of publication 11 December 2024,
date of current version 23 December 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3515215

SURVEY

Application of Deep Learning in Forest Fire Prediction: A Systematic Review

CESILIA MAMBILE^{ID}, (Member, IEEE), **SHUBI KAIJAGE**^{ID}, (Senior Member, IEEE),
AND JUDITH LEO, (Member, IEEE)

School of Computational and Communication Science and Engineering (CoCSE), The Nelson Mandela African Institution of Science and Technology (NM-AIST), Arusha 2331, Tanzania

Corresponding author: Cesilia Mambile (cesiliam@nm-aist.ac.tz)

This work was supported by U.K. International Development and the International Development Research Centre, Ottawa, Canada, as part of Artificial Intelligence (AI) for Development: Responsible AI, Empowering People Program (AI4D), under Grant 110470-001.

ABSTRACT Forests are among the world's most valuable ecological resources. However, they face significant threats from Forest Fires (FFs), causing environmental damage and impacting wildlife and economies. The increasing global occurrence of FFs has created an urgent need for more accurate prediction methods. Traditional FF prediction approaches, reliant on meteorological data and human expertise, are often limited in accuracy and scalability. Deep Learning (DL) offers a promising solution for enhancing prediction capabilities. This systematic review evaluated various DL techniques for FF prediction, analyzing their methodologies, effectiveness, and challenges. Covering studies published between January 2017 and July 2024, 55 of 656 papers were selected for detailed analysis. The study revealed that Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) models are the most frequently used, with most datasets being publicly available. These findings indicate that classification models and simulation-based studies dominate the field. Commonly used metrics include accuracy, precision, recall, F1 score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Key meteorological features, such as Temperature, Humidity, and Wind speed, have been extensively studied using the Normalized Difference Vegetation Index (NDVI), Land Surface Temperature (LST), and Normalized Difference Moisture Index (NDMI), the most commonly used satellite-derived features. However, integrating human activity data remains underexplored despite its potential to improve prediction accuracy significantly. Addressing this gap could enhance the practical applicability of DL models for FF predictions. This study provides insights into the most prevalent and effective DL techniques for FF prediction and highlights areas for future research.

INDEX TERMS Deep learning, forest fire, forest fire prediction, prediction features, satellite images, wildfire.

I. INTRODUCTION

The term 'forest' refers to a large land area densely forested by trees and other plants [1], [2]. Forests are among the world's most important ecological resources. Forest ecosystems are critical for several reasons. Millions of plants and animals call the forests to their homes. Moreover, they assist in regulating the climate, filtering air and water, and providing various valuable resources [3], [4]. Despite their value,

forests face numerous threats, such as destruction, forest fires (FFs), and deforestation [5], [6]. Approximately 67,000 FFs occur annually worldwide, destroying approximately 7 million acres (approximately 2.83 million hectares), which is a substantial loss, considering the critical role of forests in carbon storage, biodiversity, and ecosystem services [7], [8].

FFs constitute a significant concern because they release large amounts of carbon dioxide, contribute to climate change [3], [6], cause soil erosion, reduce soil fertility, and increase sedimentation in nearby rivers and streams [6], [10]. These fires are complex events that often result from natural

The associate editor coordinating the review of this manuscript and approving it for publication was Li Zhang^{ID}.

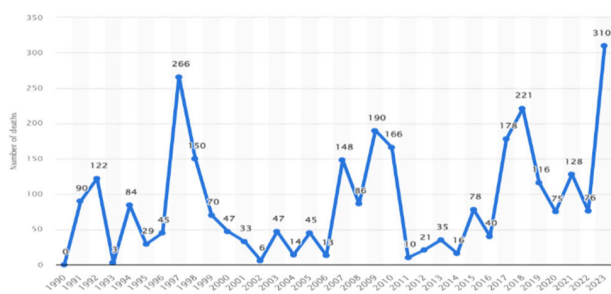


FIGURE 1. Global number of deaths caused by forest fires 1990-2023 [9].

causes, such as lightning strikes and volcanic eruptions, as well as human-induced factors, such as land clearing, charcoal burners [11], cigarette disposal, hunting, arsonists, smoking bees, and motorists [12]. Under conducive environmental conditions, dry vegetation, prolonged drought, high temperatures, and strong winds also significantly contribute to the ignition and spread of FFs, making it difficult to promptly contain and extinguish FFs [13], [14].

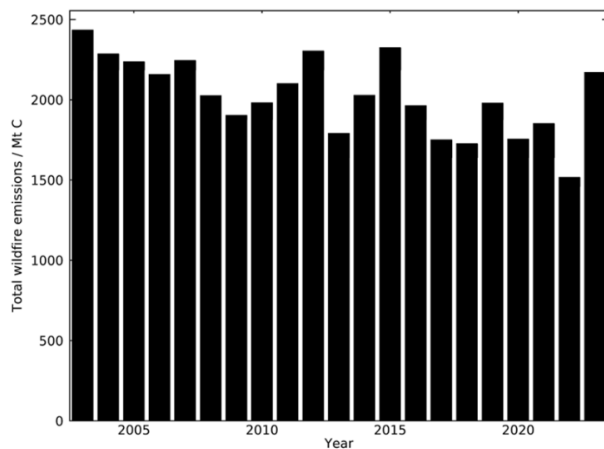


FIGURE 2. Annual global total FF emission [15].

The increasing global occurrence and severity of FFs have prompted an urgent need for effective prediction and management strategies [16], [17]. During 2023, FFs had resulted in 310 deaths. Figure 1 shows the statistics of people who have died of FFs worldwide. According to [15], 2023, global fires are expected to generate 2,170 million carbon emissions. Figure 2 shows the carbon emissions statistics for the FFs worldwide. This year (2024) was reported to have intense FF activity worldwide, with Canada experiencing its highest carbon emissions, about 22% of the total emissions, and Hawaii experiencing 115 fatalities, 26 Chilean deaths, and 1,000 injuries [9], [15]. Africa recorded a net loss of approximately 3.9 million hectares per year between 2010 and 2020. However, in recent years, estimates have indicated that the continent has lost approximately 20.5 million hectares of

tree cover annually [18]. This figure highlights the ongoing challenges faced by the FFs.

In response to combating FFs and their effects, several strategies and mechanisms were employed to prevent FFs, including traditional FF management systems such as public awareness and education [2], fire watch towers, patrols, controlled burning, early warning systems, firebreaks, firefighting equipment, participatory forest management, monitoring and surveillance, and enforcing laws and regulations [19]. Moreover, wireless sensor network-based solutions, Internet of Things (IoT) security systems [20], and meteorological models can predict fire-weather conditions. However, many of these traditional FF prediction methods are not result-oriented because they are too manual and focus on only a few factors that affect burning [21], [22]. Meteorological models have limitations in terms of accuracy and scalability [23], [24], whereas wireless sensor networks (WSNs) face many challenges, including network delay, low bandwidth, low power, transmission range, short battery lifespan, and high costs [20]. Traditional FF prediction methods based on meteorological data and human expertise have limited accuracy and scalability [25], [26].

Recent advancements in deep learning (DL) offer promising possibilities for addressing challenges in FF prediction, providing valuable insights into early warning systems, resource allocation, and mitigation planning [27], [28]. DL techniques are adept at processing large datasets and recognizing complex patterns, thereby enabling more accurate predictions of FF occurrence and behavior [17], [29], [30]. Effective FF prediction is essential for minimizing environmental damage, preserving biodiversity, reducing economic losses, and safeguarding human lives.

This systematic review consolidates knowledge from various DL models applied to FF prediction, evaluating their methodologies and effectiveness while identifying key challenges and future directions. Analyzing studies published between January 2017 and July 2024, the review provides insights into the most prevalent and effective DL techniques, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, commonly used for tasks like analyzing satellite imagery and time-series prediction. However, a notable gap remains in integrating human activity data, such as land-use changes, which could further enhance prediction accuracy and real-world applicability.

Additionally, despite the success of DL models in specific applications, the scalability and generalizability of these approaches across different geographical regions and fire conditions remain underexplored. This review bridges these gaps by evaluating studies across interdisciplinary domains, including AI, remote sensing, and environmental science, to present a comprehensive analysis of how DL models perform in FF prediction. Moreover, the review highlights current methodological gaps and provides suggestions for future research directions aimed at improving predictive accuracy and practical application.

The remainder of this paper is organized as follows: Section two provides a terminological background and an overview of the technology employed. Section three explores various DL techniques for FF prediction. Section four describes the methodology used in this literature review. Section five presents the results of the study. Section six further discusses the results of this study.

II. BACKGROUND

A. ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) is the simulation of human intelligence processes using machines, particularly computer systems [31]. Specific applications of AI include Machine Learning (ML), expert systems, natural language processing, speech recognition, machine vision, anomaly detection, and robotics [32]. The concept of AI dates back to the mid-20th century, with early developments focusing on creating machines capable of performing tasks that are considered intelligent if performed by humans [33].

The historical utilization of AI began with simple problem-solving and symbolic reasoning tasks [34], [35]. Over the decades, AI has evolved, incorporating more complex algorithms and computational power, enabling the development of advanced systems capable of learning and decision-making. The initial AI research in the 1950s and the 1960s focused on general problem-solving and game-playing, laying the groundwork for further advancement [33]. The introduction of expert systems in the 1970s and the 1980s marked significant progress, where systems could mimic human decision-making in specialized domains [36].

Early AI applications involved the use of expert systems to predict fire behavior and manage firefighting strategies in the context of fire studies [33], [36]. These systems utilize rule-based algorithms to process input data, such as weather conditions, vegetation type, and topography, to estimate fire spread and intensity. As computational capabilities have advanced, AI methods have begun to incorporate more sophisticated models that leverage historical fire data and real-time environmental input [37], [38].

The application of AI in fire prediction began with the implementation of decision trees and support vector machines (SVMs) [39], [40]. Initially, decision trees, with their hierarchical structure, provided an intuitive method for classifying fire-prone areas based on observable patterns within the data [39], [41]. In parallel, SVMs offered capabilities for binary classification tasks, such as distinguishing between high and low fire-risk zones, by leveraging the concept of hyperplanes to separate different classes within a multidimensional space [39], [40]. Collectively, these early models laid the groundwork for subsequent advancements by demonstrating the potential of AI in handling complex environmental datasets.

Building on this foundation, neural networks emerged as a significant advancement in the AI toolkit for fire prediction [41], [42]. Specifically, early neural networks,

particularly feedforward networks, were instrumental in modeling the non-linear relationships inherent in fire-related data [40]. Through their multi-layered architecture, these networks enabled more sophisticated analysis by capturing interactions between various predictive features, such as temperature, humidity, and vegetation type [43]. Thus, the introduction of neural networks marked a pivotal shift towards more dynamic and flexible modeling approaches in fire prediction.

As AI techniques continued to evolve, the advent of ensemble methods, such as Random Forests and Gradient Boosting Machines (GBMs), represented a further leap forward [40]. These methods enhanced predictive accuracy by combining the outputs of multiple models to form a consensus prediction [44]. Importantly, the ensemble approach mitigated the risk of overfitting and improved generalization across different datasets and environmental conditions [45].

Finally, the advancements in AI techniques culminated in the transition to DL models, which now dominate the field of fire prediction.

B. MACHINE LEARNING

Machine learning (ML) is a subset of AI that provides methods and algorithms that enable AI systems to learn from data [46], [47], thereby enhancing their effectiveness and adaptability. ML focuses on understanding and leveraging patterns in data and reshaping industries by redefining human interactions with technology [48]. ML algorithms perform various tasks, including classification, regression, clustering, dimensionality reduction, association rule learning, model-free learning, model-based learning, label propagation, and self-training [49].

ML techniques, such as supervised, unsupervised, and reinforcement learning, have been applied in various domains, including finance, healthcare, manufacturing, transportation, natural language processing, and computer vision [50], [51]. In the context of FF prediction, ML models analyze meteorological and environmental data to predict fire risk and behavior [51].

C. DL IN FF PREDICTION

DL, a subset of ML, is characterized by its ability to handle large, unstructured datasets, such as satellite images and video feeds, making it particularly suited for complex tasks, such as FF prediction [29], [52]. DL techniques, such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs), have revolutionized the field by enabling the processing of large datasets and the recognition of complex patterns [53], [54].

DL models can predict the likelihood of fire ignition and its progression under various conditions by analyzing data from previous fires [16]. The most common DL techniques for FF prediction include classification models, which categorize data into predefined classes, and regression models,

which predict continuous outcomes, such as fire size and spread rate [55]. Object detection models have also been employed to identify fire locations and boundaries using satellite imagery [56].

DL approaches have shown significant promise for improving the accuracy of FF predictions [57]. For example, CNNs can effectively analyze satellite images to detect fire-prone areas and active fires [53], [58]. RNNs, particularly Long Short-Term Memory (LSTM) networks, are used for time-series forecasting to predict the fire spread based on historical data [59]. GANs are utilized to generate synthetic fire data and aid in the training of the predictive models [60], [61]. However, challenges such as data scarcity, feature selection, and model interpretability remain critical areas for further research [53], [57].

There are various DL algorithms; among the most popular are Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs).

1) CONVOLUTIONAL NEURAL NETWORKS (CNNs)

CNNs are DL models that are well-suited for image and spatial data processing [41], [62]. CNNs are designed to automatically and adaptively learn the spatial hierarchies of features from input images. A typical CNN architecture comprises several layers, including convolutional, pooling, and fully connected layers [62], [63], as shown in Figure 3.

In the convolutional layer, the network applies filters (kernels) to the input image and performs convolution operations to produce the feature maps [63], [64]. These feature maps highlight various aspects of the input image, such as edges, textures, and shapes. Pooling layers then downsample the feature maps, reducing their dimensionality and computational complexity while retaining the most critical information [22], [65]. The fully connected layers at the network end use the extracted features to perform the classification or regression tasks [64].

CNNs are particularly effective for FF prediction because they can analyze satellite imagery and detect fire-prone areas and active fires [66], [67]. CNNs are multispectral satellite images that can be used to identify the specific patterns and anomalies associated with fire outbreaks. These models can also classify different types of land cover, assess fire damage, and monitor vegetation health [68].

Several studies have demonstrated the effectiveness of CNNs for FF prediction and management. For example, a study by [57] developed an improved dynamic Convolutional Neural Network (DCN_Fire) to predict FFs by analyzing UAV images of flames and smoke. The model achieved a verification accuracy of 98.3%, thereby highlighting its high efficacy for identifying active fires for timely intervention [57]. Another study [69] employed a CNN using geostationary satellite data for FF prediction with a reduced detection latency. The model achieved an overall accuracy of 98%, demonstrating its effectiveness in providing timely

alerts for early intervention [69]. Additionally, a study by Yu et al. [70] employed a CA-based CCNN with active learning optimization to predict wildfire susceptibility, the model integrates meteorological and topographical factors. The model achieved an accuracy of 91.7% and an AUC of 0.9487, demonstrating superior performance compared with traditional methods [70].

CNN plays a crucial role in post-fire analysis. Comparing pre- and post-fire satellite images, CNNs can assess the extent of fire damage and assist in recovery planning [55]. For instance, a study by Monaco [55] use CNNs to evaluate burn severity and vegetation regrowth after FFs provided valuable insights into forest restoration.

There are three types of CNNs. 1D CNNs are used for analyzing sequential data with one-dimensional inputs, such as time-series data on weather conditions [71]. 2D CNNs are commonly used for image processing tasks, including satellite imagery analysis for detecting and monitoring FFs [68], [71]. 3D CNNs are utilized for volumetric data analysis and are applicable for modeling fire behavior in three-dimensional space [24].

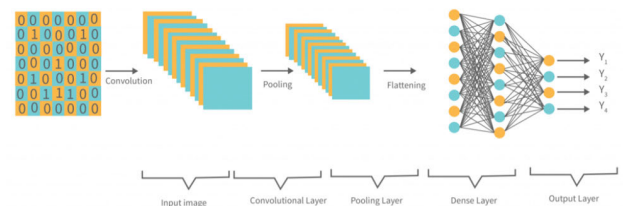


FIGURE 3. An architecture diagram of CNN [72].

2) RECURRENT NEURAL NETWORK (RNN)

RNNs are DL models that are used for sequential data processing [73]. Unlike traditional feedforward neural networks, RNNs have connections that form directed cycles, thereby allowing them to maintain memory of previous inputs [74]. RNNs are particularly suited to tasks that involve time series data and sequential dependencies [59], [75].

The fundamental building block of an RNN is the recurrent cell [73], as shown in Figure 4. The cell processed one element of the input sequence at a time [66]. The cell maintains a hidden state that captures information from previous time steps, enabling the network to model the temporal dependencies [59], [71]. RNNs can be extended to more advanced cells, such as Long Short-Term Memory (LSTM) and gated recurrent units (GRU) [76].

Long Short-Term Memory (LSTM): LSTM networks are a type of RNN designed to capture the long-term dependencies in sequential data [59]. Unlike standard RNNs, LSTMs can effectively remember important information over long periods while avoiding issues such as vanishing gradients [13], [77]. LSTMs are particularly useful in FF prediction for analyzing time-series data such as historical weather conditions and fire occurrences to forecast future fire events [75].

Gated Recurrent Unit (GRU): GRUs are another type of RNN similar to LSTMs but with a simplified architecture [75]. GRUs use gating mechanisms to control the flow of information, making them computationally more efficient while still capturing dependencies in the sequential data [73]. GRUs are employed in FF prediction for tasks that require modeling of temporal dependencies, such as predicting fire spread based on historical data [75], [76].

In the context of FF prediction, RNNs were employed to analyze time-series data related to environmental conditions and fire occurrences [76]. By leveraging historical data on temperature, humidity, wind speed, and vegetation indices, RNNs can be used to predict future fire events and their progressions [73], [74]. These models are particularly useful for forecasting fire spread, identifying potential ignition points, and assessing the fire risk over time [75].

Several studies have highlighted the effectiveness of RNNs in predicting FF. For instance, one study by Natekar et al. [59] employed LSTM networks to predict FF using meteorological features. The model achieved a high accuracy of 94.77% using multiple time steps and feature normalization, thereby aiding efficient resource allocation and evacuation planning. Another study by Zhao et al. [76] used GRU networks to detect early-stage FF using the GOES-R satellite time-series data. This GRU-based approach demonstrated superior performance in the early detection and precise location identification of FF compared with traditional methods, proving it to be a reliable FF risk assessment and management tool.

RNNs also play a significant role in integrating multisource data for a comprehensive FF prediction [73]. For example, one study by Li et al. [78] combined AUV images, weather forecasts, and historical fire records using an LSTM-based model to predict the occurrence and spread of fire. The integrated approach improved prediction accuracy and provided actionable insights for fire management agencies [79].

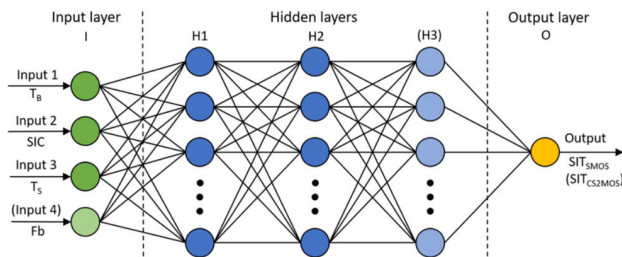


FIGURE 4. An architecture diagram of RNN [80].

3) GENERATIVE ADVERSARIAL NETWORKS (GANs)

Generative Adversarial Networks (GANs) are a class of DL models introduced by Goodfellow et al. [81]. GANs consist of two neural networks: a generator and a discriminator [82], as shown in Figure 5. These two networks were simultaneously trained in a process that can be regarded as

a game [83], [84]. The generator creates fake data samples, whereas the discriminator evaluates their authenticity by distinguishing between the real and fake samples [85]. The generator aims to produce data that is indistinguishable from real data, whereas the discriminator seeks to identify real and fake data correctly [82].

The generator begins with random noise and transforms it into a data sample [84], [86]. It then passes this sample to the discriminator, which compares it with the actual data samples and provides feedback [85]. Based on this feedback, the generator improves the data generation process [82]. Over time, the generator becomes proficient in creating realistic data, and the discriminator becomes better at detecting fake data. This adversarial training process continues until the generated data are nearly indistinguishable from the real data [84], [85].

In the context of the FF prediction, GANs generate synthetic datasets that are useful for training other DL models [87]. This is particularly valuable in scenarios where real-world data are scarce or difficult to obtain [88]. For example, GANs can create realistic fire-spread patterns based on various environmental conditions, thereby providing additional training data for predictive models [87].

Several studies have explored the use of GANs to predict and manage FF. One notable application is the generation of synthetic satellite images that mimic the actual fire conditions [89], [90]. These synthetic images can augment training datasets for convolutional neural networks (CNNs) that detect and predict FFs [87]. Another application is data augmentation, in which GANs generate varied fire-spread scenarios based on different weather conditions, terrain, and vegetation types [90]. This helps train more robust models that can generalize better to real-world conditions.

For instance, a study by Bakhshipour et al. [88] used GANs-generated synthetic time-series data that were applied to combined sewer flow prediction. These were then used to train a predictive model for fire spread. The study demonstrated that including GAN-generated data improved the accuracy of the predictive model compared to training on real data alone. Another study by Dung and Kim [87], employed GANs to generate high-resolution fire images with controllable attributes to predict FFs, aids in better model training and validation [88], [90].

Despite their potential, GANs also face challenges such as mode collapse [91], where the generator produces limited variations of data samples [92], and the difficulty of training GANs owing to their adversarial nature [90]. However, ongoing research has focused on improving the GAN architectures and training techniques to overcome these limitations.

4) OTHER ARCHITECTURES

Multilayer Perceptions (MLPs): MLPs are a fundamental type of artificial neural network that is widely used for various predictive modeling tasks owing to their versatility and simplicity [10], [29]. They consist of multiple layers of perceptrons (neurons), making them a form of deep neural

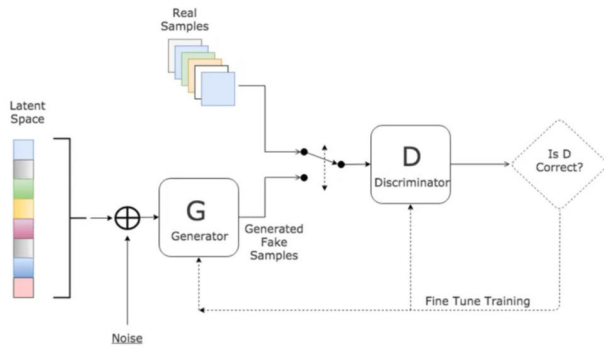


FIGURE 5. An architecture diagram of GAN [61].

network [93]. Each layer was fully connected to the next layer, allowing the MLPs to learn complex data representations [94], [95]. In FF prediction, MLPs can be used for risk assessment and regression analysis, where the goal is to predict continuous outcomes such as fire size and spread rate [96].

Multilayer perceptrons (MLPs) are valuable tools for FF prediction because of their versatility in handling various predictive tasks [29]. MLPs identify different types of vegetation and land cover, helping pinpoint fire-prone areas by distinguishing between healthy and dry vegetation [10], [93]. For regression tasks, MLPs predict the likelihood of fire occurrence, potential fire spread rates, and estimated fire size by using historical weather data and vegetation indices [96].

MLPs excel in time-series forecasting by analyzing historical weather patterns to predict future fire risks and trends, which aids in early warning systems [95]. They are also effective in anomaly detection and in identifying unusual patterns in environmental data that may indicate the onset of a fire, thus enabling early detection and intervention [94].

In pattern recognition, MLPs analyze satellite images and remote sensing data to identify fire activity indicators such as burn scars and smoke plumes [29]. They also approximate the complex relationships between environmental factors and fire behavior, helping predict the spread of fires under different conditions [95].

MLPs reduce data complexity by extracting key features from large datasets, thus simplifying analysis and visualization [10]. They also supported recommendation systems and suggested optimal fire prevention and mitigation strategies based on historical data and current conditions [96].

U-net: U-Net is a specialized convolutional network architecture designed for image segmentation [55], [97]. It features an encoder-decoder structure, where the encoder captures contextual information, and the decoder restores the spatial resolution for precise segmentation [62]. U-net is particularly effective in FF prediction, delineating fire boundaries in satellite imagery, and allowing for accurate mapping of affected areas [98]. U-net is highly effective in FF prediction and management because of its ability to capture both the context and the precise localization of features [99].

A key application of U-net is fire boundary delineation, which segments satellite images to identify and map the active fire boundaries [97], [98]. This helps to assess fire damage and plan containment strategies. Additionally, U-net was used for burned-area assessment by processing before-and-after satellite images to determine the total area burned, aid damage assessment, and recovery planning [99].

U-Net also monitors vegetation health by segmenting images to identify vegetation types and assess their health. This helps to detect stressed or damaged vegetation, indicating areas at a higher risk of future fires [100]. Furthermore, U-net detects fire-prone areas by identifying features, such as dry vegetation and high fuel load, thereby helping to create risk maps for targeted prevention measures [25].

In fire-spread modeling, U-net segment images are used to predict the direction and speed of fire spread by analyzing sequential satellite images [27]. This capability is crucial for evacuation planning and effectively deploying firefighting resources. Moreover, U-net integrates real-time monitoring systems for continuous analysis, providing up-to-date information to emergency response teams and decision-makers [101].

FireYolo: FireYolo is an adaptation of the You Only Look Once (YOLO) architecture specifically designed for real-time fire detection [102]. By leveraging the strengths of the original YOLO framework, FireYolo can quickly identify fire instances from images or video feeds, which is crucial for early detection and a rapid response [103]. Unlike traditional object-detection models, FireYolo performs detection in a single pass, accelerating the process significantly [104].

FireYolo used advanced techniques to improve accuracy and precision, incorporating a robust loss function and optimized anchor boxes to minimize false positives and negatives [105]. It employs a multiscale detection approach that allows for the detection of fires of varying sizes and resolutions across diverse landscapes and environmental conditions [106].

Designed to be lightweight and efficient, FireYolo requires less computational resources, making it suitable for deployment in edge devices and drones [106]. This enables on-site fire detection and monitoring without relying on a centralized computing infrastructure [102], [103]. The end-to-end training approach ensures that all parts of the model are optimized to achieve better performance.

FireYolo can be seamlessly integrated with other monitoring and alert systems, thereby enhancing the overall fire-management infrastructure [105]. Built on open-source YOLO architecture, it is customizable and extendable, allowing researchers and practitioners to tailor it according to specific needs [106]. These features make FireYolo a powerful FF detection and monitoring tool in various scenarios.

D. DATASETS

DL uses a dataset to train, validate, and test models depending on the application and can comprise various data types,

including images, text, audio, and numerical values [56], [107]. For effective model training, larger datasets generally provide more information, aid in training more accurate models, and require additional computational resources [108]. High-quality datasets are essential for the training of effective models. This includes aspects such as relevance of the data, minimal errors or noise, and balanced representation (particularly in classification tasks) [56], [108]. Many DL models require labeled datasets, tagging each input point with an output label. The accuracy and consistency of labels are crucial for supervised learning [109].

Datasets can be obtained from various sources, such as public datasets from different satellite platforms, Kaggle, the UCI ML Repository, and Google Dataset Search [25], [97]. Proprietary Datasets are collected and used by individual organizations for specific tasks [21], [110]. Generated Datasets are created using simulations or algorithms, particularly in scenarios where accurate data are scarce or difficult to collect [60], [66].

The reviewed studies on FF prediction using DL models have employed diverse datasets from different sources, reflecting this research domain's complexity and interdisciplinary nature.

For example, in this review, several studies have utilized well-known datasets such as the Moderate Resolution Imaging Spectroradiometer (MODIS) satellite products, which have been extensively employed for their reliability and comprehensive coverage. A study by Kang et al. [69], utilized MODIS satellite data to develop simple CCN for FF prediction. Another study by Kondylatos et al. [27] utilized MODIS satellite images to develop LSTM and ConvLSTM for wildfire-danger prediction. Also, a study by Zhang et al. [71] used MODIS satellite data to develop a CNN model for global wildfire susceptibility. The Canadian Forest Fire Weather Index (FWI) dataset was used in [111] and [112] to develop LSTM models to predict FFs, reflecting its importance in integrating weather-related variables essential for predicting fire behavior.

Landsat-8 satellite imagery is frequently utilized for its high-resolution capabilities, which enhance the accuracy of DL models in identifying and predicting FFs. For example, several studies [43], [101], [109], and [113] utilized Landsat-8 satellite images to develop different DL models, such as FU-NetCastV2, U-Net, ABi-LSTM, and BiLSTM+Attention, for FF prediction and detection. Additionally, specialized datasets, such as UAV images from Guangdong Longshan and Jiangmen Sihui Forest farms, were used by Zheng et al. [57] to validate the improved DCNN models.

FireCube and NDWS datasets were employed in [17] and [24] to develop ConvLSTM and 2D/3D CNN for wildfire danger forecasting, demonstrating the utility of integrating multiple data types to improve model robustness [17], [24]. One study [10] utilized the FF dataset created by Cortez and Marais to develop MLP models for FF prediction. In another study [43], Lidar and Spot-6 images and Australian

meteorological and census data were used to develop a Deep NN and MLP for FF prediction.

Similarly, a diverse array of models and datasets were explored, including the FU-NetCastV2 model, which utilized GeoMAC data [30], and datasets like the Sentinel-2 satellite imagery utilized in [55] to develop DS-AttU, DS-Unet, and DS-UNet++ for wildfire severity prediction and the historical wildfire data from GLOBAL FOREST WATCH in Pakistan were utilized in [26] to develop LSTM to forecast deforestation using fire alerts. The Google Earth Engine (GEE) was employed [100] for residual U-Net models, whereas Himawari-8 satellite data were used for WLF-UNet models [25].

The SeasFire cube dataset was another vital data source used in the U-Net++ models [17], and Landsat and NOAA data were used in the FireCast models [114]. Datasets such as the Tropical Rainfall Measuring Mission (TRMM), Digital Elevation Model (DEM), land cover, fire mask, Normalized Difference Vegetation Index (NDVI), and Normalized Difference Water Index (NDWI) were used by Santopaolo et al. [62] to develop a CNN model for FF risk prediction.

Moreover, a study by Kanwal et al. [68] utilized other significant datasets called the FIRMS dataset to develop a CNN (1D and 2D) for wildfire mapping and prediction assessment, and the FARSITE simulated dataset used by Li et al. [66] to develop a Mixed-Input CNN with Attention (MI-AM-CNN) for FF prediction. Another study by Marjani et al. [58] included datasets from British Columbia, Alaska, and Nova Scotia to develop FirePred for a hybrid multitemporal convolutional neural network model for wildfire spread prediction. The daily ignition dataset from Yunnan Province was used [115] for the 3D CNN-ConvLSTM models.

In addition, the review covered models using the UC Irvine ML Repository and Satellite Imagery Dataset from the Open Government Portal [116] and the Montesinho Natural Park dataset for the LBFFPS models [117], [118]. Next-day wildfire-spread datasets have been utilized in multiple studies, demonstrating their importance in short-term FF predictions [21], [53], [65]. In addition, datasets from the Geospatial Data Cloud and National Earth System Science Data Center were used in the CA-based CCNN models [70], whereas the Copernicus, Visible Infrared Imaging Radiometer Suite (VIIRS), ERA-5, and ALWB were used in various models as demonstrated in [27], [53], and [119].

The review of the datasets highlights the dynamic and evolving nature of FF prediction research with continuous efforts to refine data integration and model accuracy.

E. SATELLITE IMAGES

Satellite images are pictures of Earth or other planets captured by imaging satellites operated by governments and businesses worldwide [120]. Unlike traditional photographs, satellite images use remote sensing technology to collect data that can be processed into images that visually represent Earth's surface and atmosphere [121].

Satellite imagery is crucial for predicting and managing FFs. Monitoring large areas of forested land from space provides invaluable data for detecting active fires, assessing fire damage [122], and predicting fire behavior. Different satellite platforms offer unique capabilities that significantly contribute to these efforts [123].

For instance, Landsat satellites, managed by the US Geological Survey (USGS), provide high-resolution optical and thermal imagery of the Earth's surface [124]. Landsat data are essential for mapping burned areas, monitoring vegetation health, and identifying changes in land cover, which may indicate increased fire risk [125]. The high spatial resolution and availability of historical data make Landsat invaluable for real-time monitoring and long-term fire trend analysis [126].

Moderate Resolution Imaging Spectroradiometer (MODIS) sensors onboard NASA's Terra and Aqua satellites capture data in multiple spectral bands [127], providing daily global coverage ideal for real-time fire detection and monitoring [113], [128]. Thermal bands are particularly useful for identifying active fire fronts and hotspots, supporting early warning systems, and tracking fire progression over time [128], [129].

The Visible Infrared Imaging Radiometer Suite (VIIRS), onboard the Suomi NPP and NOAA-20 satellites, offers a higher spatial resolution than MODIS and captures data at visible and infrared wavelengths [120]. VIIRS is valuable for detecting smaller fires and continuously monitoring fire-prone areas [120], [121].

Sentinel-2, part of the European Space Agency's Copernicus program [130], provides high-resolution optical imagery in multiple spectral bands. Sentinel-2 data were used for vegetation monitoring, land cover classification, and detection of changes in forest health [99], [131]. Its frequent revisit time and high spatial resolution make it an excellent resource for assessing fire damage and for planning recovery [132], [133].

Himawari-8, operated by the Japan Meteorological Agency [134], offers high-frequency observations focused on the Asia-Pacific region, providing imagery every 10 minutes, which is critical for tracking the rapid progression of FFs [135]. Himawari-8 data support real-time fire monitoring and help to understand diurnal variations in fire activity [135].

GeoMAC (Geospatial Multi-Agency Coordination) is an online mapping application providing near-real-time FF information to the public [136]. It integrates data from multiple sources, including satellite imagery and ground observations, to offer comprehensive situational awareness of ongoing fire events, aiding emergency responders and the public in making informed decisions during fire emergencies [129].

(Geostationary Operational Environmental Satellites (GOES) provide continuous imagery of the Americas [137] and are instrumental in monitoring weather conditions that influence fire behavior, such as wind patterns, temperature, and humidity. GOES data are used to detect and assess the

meteorological conditions that may exacerbate the spread of fires [114].

The thermal and infrared capabilities of satellites such as MODIS and VIIRS enable real-time fire detection and provide authorities with early warnings [53]. Monitoring vegetation health and land cover changes is another critical application of high-resolution optical imagery from Landsat and Sentinel-2 to detect areas of stressed or dead vegetation that are more susceptible to ignition [132], [133]. Satellites also play a crucial role in post-fire assessment, helping to map the extent of burned areas and evaluate the severity of damage [138].

F. EVALUATION METRICS

In DL, evaluation metrics are essential tools used to measure the performance of models [128]. These metrics vary depending on the model type and nature of the data [139]. The standard metrics for classification models include accuracy, precision, recall, F1-score, mean average precision, and average precision [128]. Accuracy measures the proportion of correct predictions among total predictions, making it suitable for balanced datasets [139]. Precision and recall are essential in imbalanced datasets; precision focuses on the accuracy of optimistic predictions, whereas recall measures the model's ability to identify all relevant instances [128].

For regression models, metrics such as the Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared are commonly used [57]. MAE and MSE measure the difference between the predicted and actual values, with MAE providing a linear measure and MSE offering quadratic scoring of errors [128]. The RMSE, which is the square root of MSE, provides a more detailed view of the error distribution [139]. R-squared indicates the proportion of variance in the dependent variable that can be predicted from the independent variables, making it a key metric in the regression analysis [20].

Another essential tool is the confusion matrix, particularly in classification problems [128], [139]. It helps to visualize a model's performance by showing true positives, false positives, and false negatives. The Area Under the Curve (AUC-ROC) was used in the binary classification to assess the model's ability to distinguish between classes [49], [66].

No single metric can evaluate the performance of a model accurately [139]. The choice of metrics depends on the specific requirements and context of the task, and multiple metrics are recommended for comprehensively understanding the performance and applicability of a model in real-world scenarios [16], [20].

G. RELEVANT FEATURES FOR FF PREDICTION

Features are fundamental elements or characteristic DL models that are used to make predictions or classifications [53]. Each feature is a measurable property or attribute of an observed phenomenon [131]. In datasets, these features can be attributes or columns, for example, representing wind

speed, wind direction, or temperature for meteorological data or distance from road for human activity data [16]. Effective FF prediction models rely on identifying and analyzing key environmental, meteorological, and human-related features influencing fire behavior.

First, meteorological features such as temperature, humidity, rainfall, and wind speed are crucial [13], [74]. Higher temperatures increase the probability of fires starting and spreading rapidly, whereas humidity levels significantly determine the moisture content of vegetation [13]. Low humidity leads to dry conditions that are conducive to fire ignition and spread, whereas high humidity suppresses fire activity [113]. The wind speed and direction are critical because strong winds can rapidly spread fires and carry embers that ignite new fires [112]. Additionally, precipitation data helps to understand moisture levels in the environment; recent rainfall can reduce fire risk, whereas prolonged dry periods can increase it [13].

Second, the environmental features are equally important [11]. Vegetation indices, such as NDVI and EVI, assess vegetation health and density and provide insights into available fire fuel [27]. Soil moisture content influences moisture availability to vegetation, with low soil moisture increasing the fire risk [58]. Topographical features, such as elevation, slope, and aspect, affect fire behavior by Kondylatos et al. [27] influencing wind patterns and fuel distribution [51]. The fires spread rapidly and were more intense on the steeper slopes.

Third, human activities increase the risk of FF, and features such as land clearing, charcoal burning, cigarette disposal, hunting, arson, smoking, and motorist activities are critical human-related features [140], [141]. Monitoring these activities helps to predict and mitigate fire occurrences caused by human intervention [142].

Finally, the inclusion of satellite imagery data enhanced the FF prediction models. High-resolution optical imagery captures visible light to identify burn scars and changes in the vegetation cover [23]. Infrared imagery helps distinguish between healthy and stressed vegetation, detect moisture content, and identify hotspots [121]. Multispectral and hyperspectral imagery provide detailed chemical and mineral information about the land surface [98], which is useful for assessing fuel types and conditions. Synthetic Aperture Radar (SAR) penetrates smoke, clouds, and darkness and monitors fire progression and post-fire assessments. Lidar offers detailed 3D information about the Earth's surface, supporting topographical assessments and modeling fire spread. Radiometer data provides temperature data to detect hotspots associated with active fires [129].

III. DL FOR FF PREDICTION TECHNIQUES

The prediction and detection of FFs have seen significant advancements through three main technological approaches: imagery-based, sensor-based, and IoT (Internet of Things) methods [20], [143]. Imagery-based detection uses fixed cameras, satellites, and drones to capture aerial images, which

are effective for early fire detection and monitoring in large areas, particularly remote locations [25], [28]. However, this method is limited by the adverse weather conditions and delayed data processing [98]. On the other hand, sensor-based detection employs environmental sensors placed throughout forests to measure data like temperature, humidity, and smoke levels [144]. This approach provides real-time data and quick detection but requires an extensive sensor network and regular maintenance [52].

IoT-based prediction techniques represent an integrated approach that combines imagery and sensor-based methods with IoT technology [145]. This involves collecting data from multiple sources, which are then analyzed using advanced algorithms and DL models for real-time monitoring, predictive analysis, and automated response [146]. While offering comprehensive and predictive insights, this method faces challenges such as complexity, cost, data security, and reliance on stable internet connectivity [16].

Each approach has its strengths and limitations. Imagery-based methods excel in broad surveillance but struggle with environmental constraints [28], whereas sensor-based techniques are precise but require dense device networks [144]. IoT-based methods merge the advantages of the previous two methods but at a higher complexity and cost [143], [145]. The choice of the method depends on various factors, including the size of the area, geographical features, and budget. Combining these techniques provides a robust and effective FF detection and prevention method.

IV. METHODOLOGY

This study systematically reviewed literature from peer-reviewed journals, conference papers, books, and book chapters published between January 2017 and July 2024. This period was selected due to significant advancements in DL, satellite imaging, and mobile computing [37], which have contributed to substantial progress in understanding and predicting FF [147]. The review captures state-of-the-art methodologies, tools, and theories that have emerged in recent years.

Data for the review were retrieved from reputable databases, including the Association for Computing Machinery (ACM) Digital Library, ScienceDirect, IEEE Xplore, Wiley Online Library, Springer Link, Scopus, Emerald, Google Scholar, and Sage Publishing. These databases were selected based on their relevance to the topics under review, alignment with the research questions, and their comprehensive bibliographic coverage. Their inclusion ensured that the review covered a wide array of high-impact journals and conferences, contributing to the thoroughness of this systematic review [148]. A review protocol from [149] was adopted in this study, as shown in Figure 6.

A. RESEARCH QUESTIONS

This study answers the following research questions.

- i. What are the most prevalent and effective DL techniques in FF prediction across different research

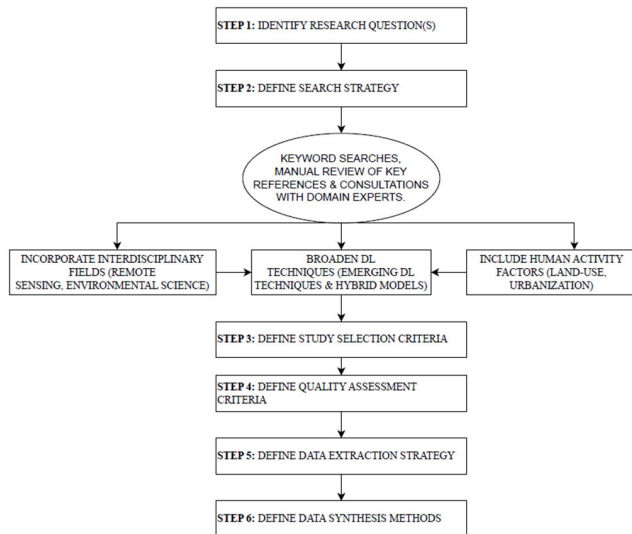


FIGURE 6. Systematic literature review protocol adopted from [149].

contexts, and what specific features enhance the accuracy and effectiveness of these models?

- ii. How does dataset quality, including the integration of human activity data and the standard metrics used, reflect the capabilities and performance of these models in FF prediction?

B. SEARCH STRATEGY

A search string was developed based on the research questions to define the scope of the studies. Boolean operators were used to combine relevant keywords: (“Deep Learning” OR “deep learning” OR “Neural Network,” OR “Neural Network” OR “Machine Learning,” OR “Machine Learning”) AND (“forest fire,” OR “wildfire,”) AND (“prediction,”) AND (“application,”) OR (“implement,”) OR (“case study,”) OR (“use case.”))

This search string was applied across multiple academic databases, including AI-focused platforms (e.g., IEEE Xplore) and interdisciplinary sources (e.g., ScienceDirect, Springer). Recognizing the limitations of relying solely on keyword-based searches, the strategy was refined to include additional terms from adjacent fields such as environmental science, remote sensing, and human activity. This expanded approach incorporated a manual review of key references and consultations with domain experts to ensure comprehensive coverage of underrepresented topics and emerging AI techniques.

The enhanced search strategy incorporated the following approaches:

- i. Inclusion of Interdisciplinary Research: Studies from remote sensing and environmental science were included to provide essential data relevant to DL models in fire prediction, including vegetation indices and climate patterns.
- ii. Broader Scope of DL Techniques: In addition to conventional DL models, emerging techniques such as

advanced CNNs and hybrid models were considered for their potential to improve prediction accuracy and manage complex datasets

- iii. Consideration of Human Activity Factors: Studies that integrated human-induced variables, such as land-use changes and urbanization, were included to account for the broader context influencing FF risks.

C. STUDY SELECTION CRITERIA

Inclusion and exclusion criteria were applied to ensure a focused and relevant selection of papers. Studies included if they:

- Were published in peer-reviewed journals, books, or conferences between 2017 and 2024.
- Addressed the application of DL techniques, such as CNNs, LSTMs, and GANs, in FF prediction.
- Used real-world datasets, including satellite imagery, meteorological data, or data representing human activities like land-use changes.
- Were written in English.

The selection process followed the PRISMA framework and was conducted in several phases to ensure a systematic approach. An initial search yielded 656 articles. Papers were screened by language, reducing the count to 642. Subsequent keyword filtering narrowed the pool further, resulting in 247 papers. A temporal restriction was then applied, leaving 159 studies. After title and abstract screening, 70 papers remained, and full-text reviews were conducted, leading to the final selection of 55 papers. The process is outlined in the PRISMA flow diagram (Figure 7).

D. QUALITY ASSESSMENT

Two independent reviewers evaluated the quality of the selected studies. Any discrepancies were resolved through discussion or by consulting a third reviewer. This approach ensured that the final pool of studies was both relevant and of high quality, accurately representing the state of DL research in FF prediction.

E. DATA SYNTHESIS

The data synthesis phase involved summarizing the findings of the 55 selected studies to answer the research questions. Key concepts were identified, conclusions were drawn about the most effective DL techniques in FF prediction, and gaps in the current methodologies.

V. RESULTS

A. THEMATIC ANALYSIS OF EMERGING THEMES IN DL AND FF PREDICTION

The selected papers revealed the following thematic areas, as shown in Figure 8. These keywords, which encompass “DL,” “machine learning,” “artificial intelligence,” “neural networks,” “convolutional neural network,” “remote sensing,” “Google Earth Engine,” “wildfires,” and “Sentinel-2,” not only define the methodological and technological toolkit

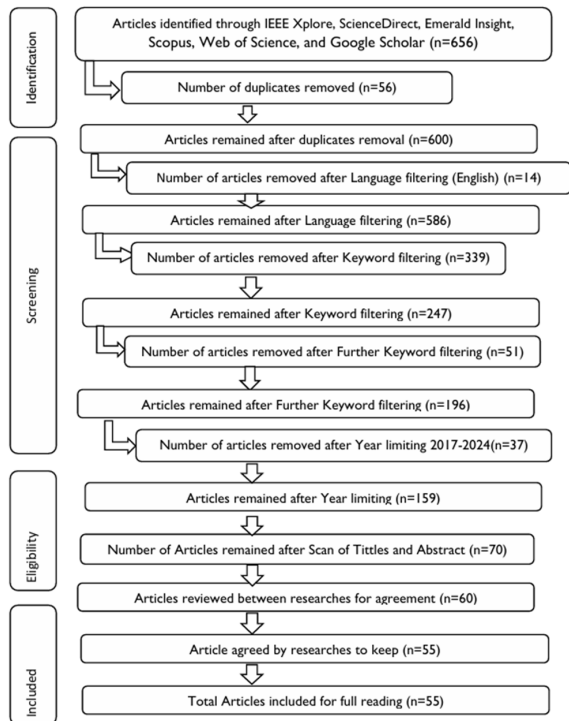


FIGURE 7. PRISMA Flow diagram of the study selection process (Source: Author's creation).

prevalent in the field but also highlight the interdisciplinary nature of current studies.

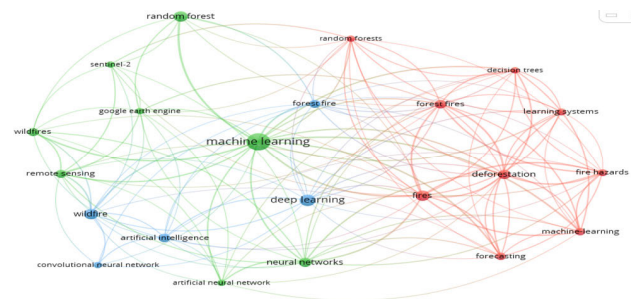


FIGURE 8. Interconnected keywords in DL and FF prediction (Source: Author's creation).

B. GLOBAL LANDSCAPE OF DL RESEARCH FOR FF PREDICTION

In the context of this DL for FF prediction, it has become evident that DL for FF prediction has significantly advanced in countries such as the United States, China, France, Italy, India, Portugal, Greece, and Canada, as shown in Table 1 below, where access to advanced computational resources, substantial investments in AI research, and the availability of diverse ecological data have enabled sophisticated predictive modeling. However, the findings of this study indicate that Africa and other developing regions have failed to catch up in

this critical area because of limited technological infrastructure, scarcity of localized datasets, and gaps in AI expertise and implementation.

TABLE 1. International distribution of DL research in FF prediction.

<i>Country</i>	Documents	Citations
<i>United States</i>	20	135
<i>India</i>	16	78
<i>China</i>	15	121
<i>Canada</i>	8	53
<i>Italy</i>	8	101
<i>France</i>	7	9
<i>Greece</i>	6	16
<i>Portugal</i>	5	40

C. TRENDS OF PUBLICATION ON DL FOR FF PREDICTION OVER THE YEARS

Figure 9 illustrates the trend of publications on DL for FF prediction from 2017 to 2024, showing a clear progression in research activity. Minimal activity was observed in the early years (2017-2018), with each year accounting for only 3.03% of the total publications, marking the initial exploration phase with foundational work and preliminary studies. During the growth phase (2019-2020), there was a clear increase, with 2019 showing an increase of 12.12% in total publications.

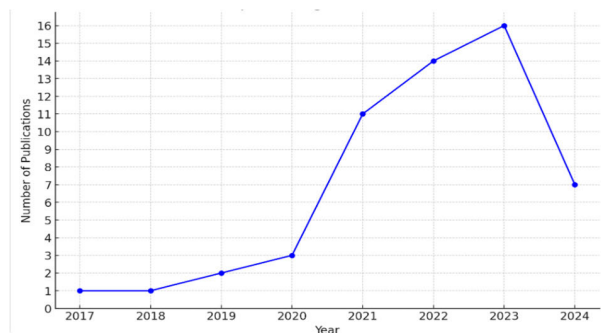


FIGURE 9. The trend of publications on DL for FF prediction over the years (Source: Author's creation).

The peak research activity (2021-2023) shows a significant surge, with 2021 accounting for 36.36% and 2022-2023 reaching the highest publication activity of 42.42%. This period indicates strong innovation, likely due to the maturation of DL models, availability of datasets, and increased computational resources.

However, despite the increase in publications and advancements in DL technologies, the decline in 2024 suggests that several challenges remain unaddressed. These issues include heterogeneity [119], quality, and consistency of the data [101]. Variability in data formats, resolutions, and collection methodologies from different sources continue to pose

significant obstacles to the development of accurate and reliable prediction models [70], [150]. This highlights the critical need for standardized protocols and improved data integration methods to ensure the reliability and applicability of DL models in real-world FF prediction scenarios. Therefore, addressing these challenges is important for further progress while the field advances.

D. DL TECHNIQUES IN FF PREDICTION

Various DL techniques have been explored for FF prediction across 55 research papers, each with applications, effectiveness, and limitations.

1) TYPE OF MODEL USED

The fact that classification models were used 56% of the time suggests that they are the primary approach for categorizing data into predefined classes. This prevalence underscores the significant role of classification models in various research contexts, likely because of their effectiveness in categorizing and differentiating between various data types or conditions. The second most frequently used model type is “Regression,” noted in 24% of papers, as shown in Figure 10. Regression models are essential for predicting continuous outcomes, such as the potential size of an FF, its duration, the rate at which it might spread, or the amount of area that could be burned. This indicates their applicability in situations requiring forecasting trends or examining the relationships between variables.

Furthermore, object detection, although less common, suggests a more specialized approach to tackling this problem. Object detection models are beneficial for FF prediction when analyzing satellite imagery or aerial photographs to identify fire locations and boundaries. The mention of “Classification and Regression” in a single paper indicates a comprehensive approach that utilizes the strengths of both methodologies to address complex research questions that benefit from discrete categorization and continuous prediction.

However, the emphasis on classification and regression models points to a potential gap in leveraging more advanced or hybrid modeling approaches. Although these models are effective, the evolving nature of FF prediction requires exploring and integrating newer methodologies [58] such as hybrid models that combine classification, regression, and object detection techniques [79], [115], which are more comprehensive and robust predictive frameworks [53]. The current reliance on traditional models may limit the innovation potential, emphasizing the need for continued research on more sophisticated models that can handle the complexity and variability inherent in FF prediction.

2) TYPE OF STUDY CONDUCTED

The study found that 68% of the studies were categorized as “Simulation,” as shown in Figure 11. The higher the frequency of simulation studies, the more preferred methods

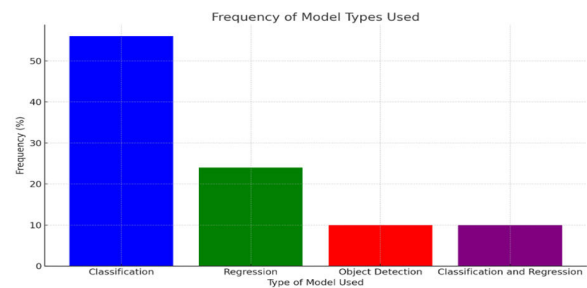


FIGURE 10. Frequency of model types used in reviewed papers (Source: Author's creation).

are computational models for testing the hypotheses and scenarios. In DL, simulations validate models when accurate word data are missing, expensive, or risky. However, a smaller proportion of experimental studies (32%) showed more constraints when conducting real-world experiments.

Studies that have relied on simulations rather than real-world testing include those conducted by Ananthi et al. [145], who conducted a simulation-based study using IoT and DL for FF prediction, and [146], who focused on simulation-based high-resolution fire-danger mapping using DL. Other examples include [114], who developed FireCast, a simulation-based model leveraging DL to predict wildfire spread, and [95], who applied Artificial Neural Networks and Genetic Algorithms to predict FF danger in simulated environments. Further simulation-based studies were conducted by Gao et al. [94], using deep U-Nets for improved wildland fire spread prediction, and [147], using a deep neural network method in simulations to predict FFs with imbalanced data, and [10] developed a simulation-based model for FF prediction using DL. In addition, [52] created an early FF detection system using wireless sensor networks and DL in a simulated setup, Naderpour et al. [29] developed a spatial deep neural network-based framework for FF risk prediction, [30] predicted the FF spread using DL models in simulated scenarios, [22] and [71] used deep neural networks for global wildfire susceptibility modeling and [27] focused on wildfire danger prediction and understanding using DL in a simulated context. Finally, [49] predicted the occurrence of FF in China using DL methods based on simulations.

However, the heavy reliance on simulations suggests a potential gap in the strength and applicability of the developed models because simulations cannot entirely replace the insights gained from real-world experiments.

3) DL MODELS

This study revealed that CNNs and LSTMs are the most frequently used models, with CNNs appearing at 48.78% and LSTMs at 12.20%, as shown in Figure 12. This dominance reflects their suitability in handling spatial and temporal data. CNNs excel in processing satellite imagery and spatial data, whereas LSTMs are well suited for time-series data, such as historical fire patterns.

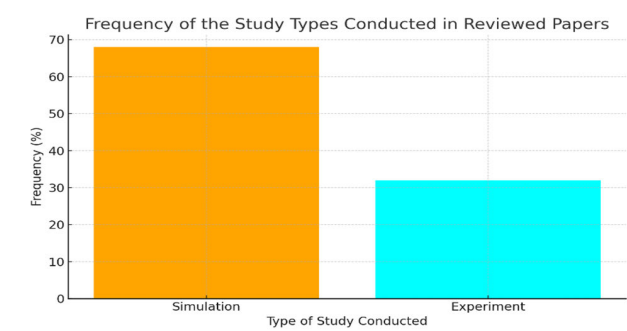


FIGURE 11. Frequency of study types in review papers (Source: Author’s creation).

A noticeable trend was the emergence of the hybrid models. The FU-NetCastV2 model, which combines the elements of CNNs and LSTMs, appears approximately 7.32% of the time. This indicates a move towards leveraging multiple architectures to improve predictive performance. This trend is reinforced by the appearance of models, such as FU-NetCastV2 and ConvLSTM, emphasizing the growing interest in hybrid approaches.

Furthermore, the field exhibits diverse experimentation, with many models appearing only once. This highlights the experimental nature of research using various architectures, such as DS-AttU, DS-Unet, DS-UNet++, WLF-UNet, Ensemble learning, AllConvNet, and Residual U-Net, to identify effective solutions for different data types and prediction scenarios.

Advanced techniques have become increasingly popular in the recent years. Models incorporating advanced techniques such as Deep CNN (Fire-Net), BiLSTM+Attention, and ensemble learning have been utilized to enhance prediction accuracy. These models aim to capture complex data patterns more effectively.

However, there is a significant need for standardization in this field. The wide variety of models and their relatively equal distributions indicates a lack of consistency and comparability across studies [56], [108]. This diversity highlights the experimental nature of several models and underscores the importance of establishing clear benchmarks and guidelines [147], [150]. Comparative studies are required to systematically evaluate the performance of these models, leading to more standardized approaches and benchmarks [56], [139]. The field would benefit from coordinated efforts to align methodologies and evaluate models against standardized criteria [108], [142]. This enhances the reliability of the FF prediction models and facilitates their practical application in diverse environmental and geographical contexts.

E. DATASETS AND FEATURES

1) OVERVIEW OF COMMONLY USED DATASETS

A review of 55 papers found that imagery data are heavily utilized, indicating a trend toward reliance on remote sensing technologies for monitoring FFs (Figure 13). Additionally,

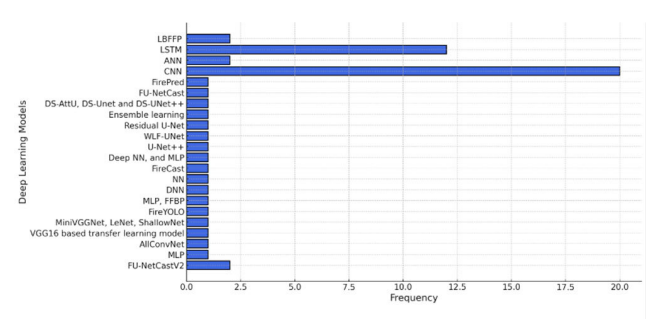


FIGURE 12. Frequency of DL models for FF prediction in reviewed papers (Source: Author’s creation).

comprehensive climate and weather data, including temperature, humidity, and wind speed, are frequently integrated, underscoring their critical role in predicting fire behavior. Datasets often combine multiple data types, such as climate, vegetation, and topographic data, signifying a holistic approach to fire analysis. Furthermore, there has been a pronounced focus on historical data and burned areas.

However, Despite the comprehensive integration of natural data, this review highlights a significant gap in including human activity data [140], [150]. While these studies emphasize natural factors, critical anthropogenic influences such as land-use changes, agricultural practices, and other human activities are often underrepresented [141]. This oversight is challenging because human activities substantially influence fire ignition and spread [34].

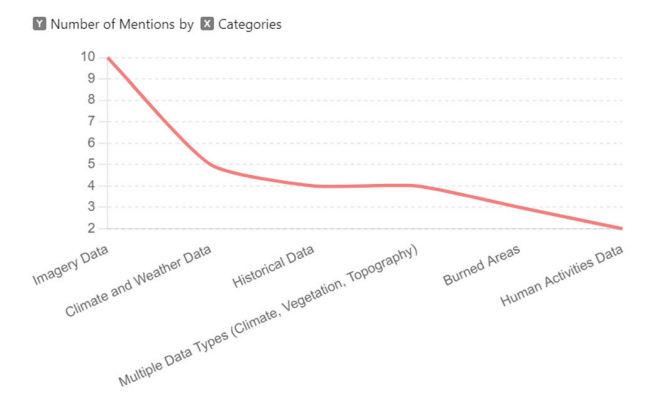


FIGURE 13. Trends in FF data from reviewed papers (Source: Author’s creation).

Furthermore, the richness of these datasets was visually encapsulated in a word cloud generated from the dataset descriptions mentioned in the reviewed papers (Figure 14). This visual representation underscores the most recurrent themes, with the size of each word reflecting its frequency and, thus, its prominence within the collective research data sources. Although a word cloud offers an intuitive grasp of a dataset’s common threads, it does not consider the specific quantitative data.

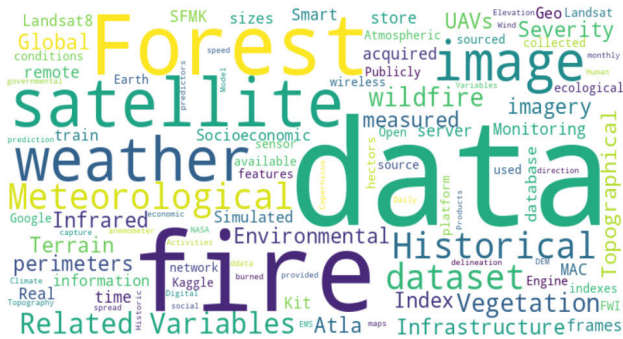


FIGURE 14. Summary of the most frequently occurring words across all dataset descriptions (Source: Author's creation).

Interestingly, the visual representation of the dataset descriptions in Figure 14 reveals a significant gap in the current body of research: under-representation of human activity data. Despite the prominence of terms related to natural factors like “satellite,” “weather,” “data,” and “forest,” critical anthropogenic factors such as “land-use changes,” “agricultural practices,” “charcoal burning,” “disposal of cigarettes,” “hunting,” “arson,” “smoking out bees,” and “urbanization” are notably absent or underemphasized. This omission is striking, given the well-documented role of human activities in fire ignition and spread, particularly in regions where human-induced fires are prevalent. Addressing this gap will lead to stronger models that can better predict and manage fire risks by considering natural and anthropogenic factors.

Moreover, the accessibility status of various datasets related to FFs and satellite imagery revealed significant trends and implications (Figure 15). The analysis indicated that 71% of the datasets are publicly available. In addition, a significant portion of the datasets (22.58%) was available upon request from the authors or respective organizations. Furthermore, a small fraction of the datasets remained inaccessible (6.45%). The findings clearly show a positive trend towards open data, with the majority of datasets being publicly accessible. This trend is key to advancing scientific research, enhancing

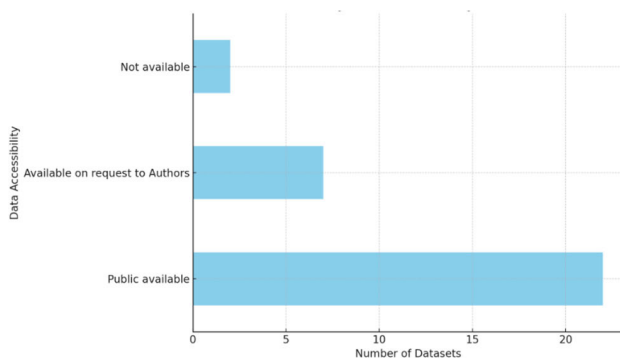


FIGURE 15. Accessibility status of various datasets used for FF prediction from reviewed papers (Source: Author's creation).

disaster preparedness and response, and supporting environmental conservation.

However, the increased availability of public datasets raises concerns regarding the quality and consistency of publicly available data [151], [152], mainly when created by people from different areas with varying specifications. The integration of diverse datasets into the FF prediction presents several challenges [108]. Data heterogeneity is a significant issue because these datasets come from various sources and are collected using different methodologies and formats [152]. For instance, satellite imagery data may have different resolutions, spectral bands, and temporal frequencies than those of meteorological or socioeconomic data [56], [108]. This variability makes seamless combinations difficult, necessitating extensive preprocessing and standardization [151].

Moreover, datasets from different regions may have specific characteristics that are not directly applicable to other areas [153], limiting their utility in broader applications even if publicly available [62]. In addition, inconsistent data can result from different measurement techniques, sensor errors, or data entry mistakes, leading to inaccurate model predictions [151]. Different datasets often have varying temporal and spatial resolution. For example, satellite data are available daily or weekly, whereas meteorological data are recorded hourly [154]. Aligning these datasets temporally and spatially is challenging; however, it is necessary to ensure that the integrated data accurately represents the conditions leading to the FFs [21]. Misalignments can result in incorrect model training and prediction, further complicating the development of accurate and reliable prediction models [110].

2) RELEVANT FEATURES FOR FF PREDICTION

The analysis of the features, mainly through the pie chart as shown in Figure 16 below, reveals a strong focus on “Temperature,” “Humidity,” and “Wind speed and direction.” The top three factors were considered the primary variables affecting FF behavior. This study discovered that these are critical inputs for DL models that aim to predict FF risk and behavior. Other factors such as “Precipitation,” “NDVI,” and “Rainfall” were also notable, and this suggests that environmental health and moisture content are significant for modeling fire risk.

However, several factors were mentioned only once, indicating that they were not considered relevant and were understudied in the context of the reviewed papers. For example, human activity-related factors such as distance from roads, human settlements, and population density have been less studied. Ignoring human-related factors can lead to an incomplete understanding of the fire risk.

Furthermore, this study identified several key features that significantly influence the accuracy of DL models in predicting FF using satellite imagery. These features encompass a range of environmental and meteorological variables that are critical for developing reliable prediction models. A detailed

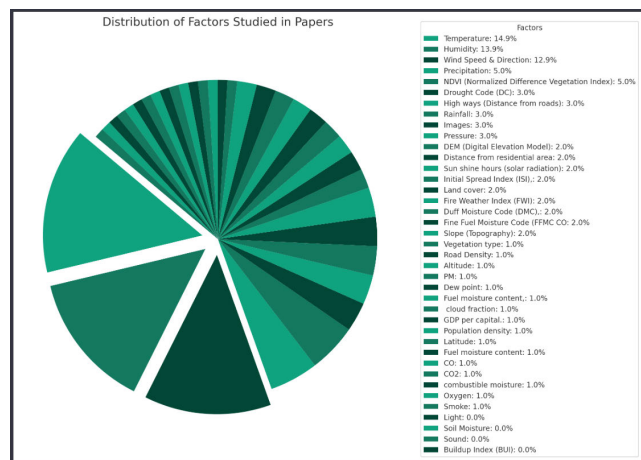


FIGURE 16. Summary of reviewed literature based on the used Features (Source: Author’s creation).

analysis and synthesis of these factors are summarized in Table 2.

TABLE 2. Summary of the features most used to predict FF using satellite images based on reviewed literature.

Feature	Description	References
Normalized Difference Vegetation Index (NDVI)	NDVI to measure vegetation health; low values may indicate potential fire fuel.	[97], [131], [150]
Land Surface Temperature (LST)	Satellite-derived temperature readings: Higher temperatures can suggest increased fire risk.	[28]
Normalized Difference Moisture Index (NDMI)	Satellite measurements of soil and vegetation moisture; dry conditions can elevate fire risk.	[62], [138]
Topography (elevation, slope, aspect)	Elevation maps from DEMs: steep slopes may lead to faster fire spread.	[56], [62]
Historical Fire Data	Locations of previous fires: Areas with frequent fires may be at higher risk.	[56], [58], [69]
Weather Data (wind speed, humidity, precipitation)	Satellite data on current weather conditions shows that high winds can exacerbate fire spread.	[118], [138], [150]
Fuel Type and Load	Information on vegetation type and density: dense, dry forests may burn more intensely.	[69], [155]
Proximity to Human Activities	Distance to roads and urban areas, which are areas closer to human activities, may have higher fire occurrence.	[27], [53]
Temporal Data	Seasonal and annual vegetation changes; drought periods can increase fire susceptibility.	[64], [101]
Spectral Bands	Infrared bands can detect heat and identify active fires and burnt areas.	[62], [64]

3) MOSTLY USED SATELLITE PLATFORMS FOR FF PREDICTION

Many satellite platforms are commonly used in this field, each with unique advantages and constraints. Figure 17

presents a summary of the most widely used satellite platforms. The results showed that Landsat program satellites were used most frequently (25%), indicating their essential role in FF prediction and environmental monitoring. The platform has a long history and provides a comprehensive data archive that helps train DL models for historical events. The MODIS and Sentinel satellites were observed at a frequency of (13%).

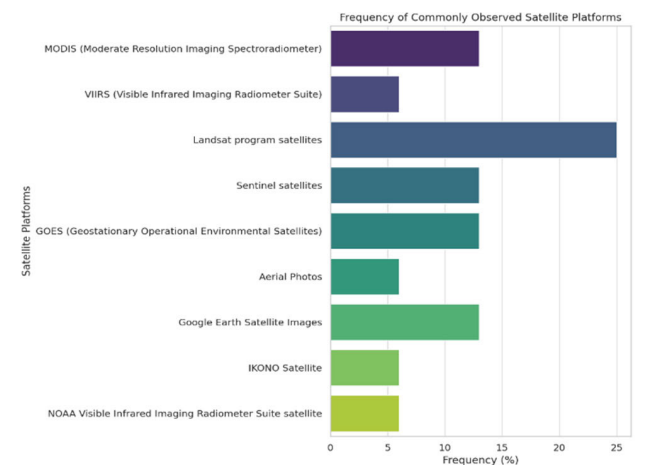


FIGURE 17. Summary of common satellite platform used for FF prediction (Source: Author’s creation).

These platforms have different characteristics. MODIS provides twice-daily global coverage, which is beneficial for near-real-time fire detection. In contrast, Sentinel provides higher-resolution images and different spectral capabilities (Sentinel-2) or rare images (Sentinel-1 -1) [98], which can be critical for penetrating smoke and cloud cover [56]. Moreover, Platforms such as VIIRS, GOES, and NOAA Visible Infrared Imaging Radiometer Suite satellites offer infrared sensing of 65 and 13%, respectively. Infrared spectroscopy is important for detecting the heat anomalies associated with fires, including smoke and cloud cover [28], [138]. In addition, aerial photos and Google Earth Satellite Images at 6% and 13%, respectively, suggest that these platforms are used to supplement satellite data, likely providing higher-resolution images that can help validate fire detection [156].

However, despite the diverse utilization of satellite platforms, the integration of data from these sources remains a significant challenge owing to the differences in resolution, spectral capabilities, and temporal coverage [108]. The reliance on multiple platforms highlights the complexity and need for comprehensive preprocessing and standardization to ensure consistency and accuracy in FF prediction models. The integration of heterogeneous data sources can lead to discrepancies and potential inaccuracies in predictions [21] Therefore, future research should focus on developing standardized data integration and validation methodologies to fully leverage each satellite platform’s strengths and improve the robustness of FF prediction models.

F. PERFORMANCE EVALUATION METRICS

This study revealed that commonly used metrics include accuracy, precision, recall, F1 score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Each of these metrics offers a unique perspective on the capability of the model to accurately predict FFs and its efficiency in reducing false positives and negatives [114]. Figure 18 summarizes the most commonly used metrics for evaluating the DL of the FF.

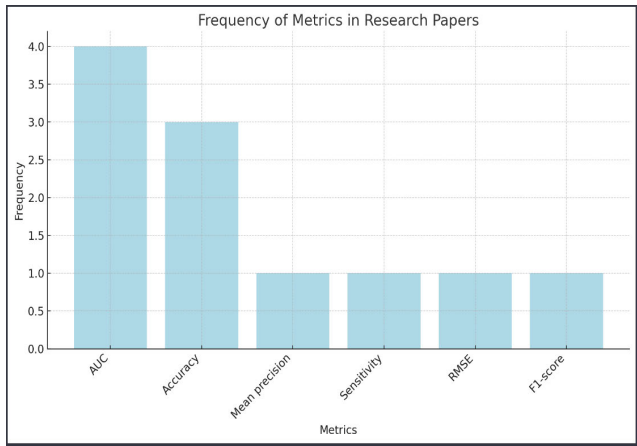


FIGURE 18. Summary of reviewed literature based on the used Metrics (Source: Author’s creation).

However, Recent research has highlighted the complexity of evaluating the FF prediction models [142]. Relying on these metrics alone may not fully capture FF prediction models’ complexities and real-world applicability. While these metrics provide valuable insights into model performance, they can sometimes overlook contextual factors such as fires’ spatial and temporal dynamics and the specific challenges associated with different geographic regions [27]. In addition, these metrics may not adequately reflect the mode’s performance in rare event scenarios, such as predicting the occurrence of FFs in regions with infrequent fire events [142]. Therefore, it is important to incorporate additional evaluation criteria and context-specific performance indicators to ensure that the DL models are accurate and practical for real-world FF management [150]. Future research should aim to develop and adopt more comprehensive evaluation frameworks that include these broader aspects to enhance the applicability and effectiveness of the FF prediction models.

G. CHALLENGES AND LIMITATIONS

1) DATA LIMITATIONS

This study discovered that one of the primary challenges in dataset selection is the availability and accessibility of high-quality data, as summarized in Table 3.

These challenges emphasize the need for more advanced and integrated approaches to data collection and analysis [56], [108]. The difficulties in obtaining and integrating

TABLE 3. Summary of challenges in dataset selection.

Dataset Challenges	Description	Reference
Data Availability and Accessibility	Obtaining high-quality, high-resolution data is challenging, especially in high-risk areas. Historical data may not capture the current dynamics of fire susceptibility or the latest land-use changes.	[17], [145]
Data Volume and Diversity	Handling large volumes and various data, such as vegetation type, climatic data, and DEM, poses integration and processing challenges. It requires sophisticated spatial analysis and logistic regression modeling.	[54], [114]
Real-Time Data Integration	The need for real-time data integration is crucial but challenging due to logistical and technical constraints. Historical datasets may not fully capture the rapidly changing conditions of FFs.	[53], [105], [156]
Data Challenges	The availability and complexity of integrating human activity data is a challenge. Such data is more complex to collect and standardize than environmental data.	[22], [23], [109]
Model Complexity	Incorporating human factors increases model complexity, requiring more sophisticated approaches and resources for model development and analysis.	[13], [57], [71]

high-quality data collection methods and infrastructure highlight the need for improved data collection methods [17]. Addressing these challenges is vital to developing robust and reliable FF prediction models.

2) CHALLENGES IN EVALUATING DL MODELS FOR FF PREDICTION

Evaluation of DL models for FF prediction presents several challenges. The study identified key difficulties such as data imbalance, temporal and spatial variability, and the need for extensive data preprocessing and interpretability. These challenges are summarized in Table 4, highlighting the various obstacles that must be overcome to develop effective DL models for FF prediction.

These challenges highlight the critical need for ongoing advancements and improvement in this field. The imbalanced nature of FF datasets, temporal and spatial variability of fires, and complexity of the phenomena underscores the necessity for innovative data augmentation techniques, more sophisticated model optimization, and hyperparameter tuning methods. In addition, the significant computational resources required for DL models and the need for real-time processing capabilities emphasize the importance of developing efficient algorithms and hardware solutions. It is important to ensure that the models generalize well to unseen data and integrate them seamlessly into the existing FF management systems for practical applications.

TABLE 4. Summary of challenges in evaluating DL models for FFs.

Evaluation Challenges	Description	Reference
Data Imbalance	The imbalanced nature of FF datasets, where non-fire events significantly outnumber actual fire occurrences, potentially leads to overestimated model accuracy and insufficient detection of actual fire events.	[110]
Temporal and Spatial Variability	FF's temporal and spatial variability impacts model generalizability. Seasonal changes, weather conditions, and geographical diversity require testing under various conditions.	[62], [98]
Data Preprocessing and Interpretability	The complex data preparation task for model training and challenges in interpreting DL models in the FF prediction context. Understanding how models make predictions is crucial for real-world applications.	[27], [55]
Complexity of FF Phenomena	FFs are complex phenomena influenced by vegetation type, meteorological conditions, topography, and human activities. Accurately modeling these factors with DL is challenging.	[49], [66]
Advancements in Technology and Data Collection	Evolving technology and methods in data collection, such as various satellite systems and sensors, add complexity to model evaluation. Models must be continually updated and evaluated against new data and methods.	[16]
Data Quality and Quantity	The limited availability of high-quality, labeled data can hinder model performance. Ensuring data accuracy and collecting sufficient data is crucial.	[22], [23]
DL Model Optimization	Optimizing DL models involves adjusting numerous parameters and architectures to achieve the best performance, which can be time-consuming and complex.	[24], [71], [142]
Hyperparameter Tuning	Selecting the optimal set of hyperparameters (e.g., learning rate, batch size) is critical for model performance but often requires extensive experimentation.	[53], [68]
Computational Resources	DL models, especially large ones, require significant computational power for training and inference, which can be a barrier for many organizations.	[54], [118]
Data Augmentation	Techniques to artificially increase the size of the training dataset are essential to improving model robustness, but they must be carefully designed to avoid introducing noise.	[69], [116]
Generalization	Ensuring that models generalize well to unseen data is challenging, particularly when training data is limited or not representative of all possible scenarios.	[64], [69]
Interpretability and Explainability	DL models often act as "black boxes," making it difficult to understand their decision-making processes. This can hinder trust and validation.	[116], [119]

TABLE 4. (Continued.) Summary of challenges in evaluating DL models for FFs.

Real-time Processing	Deploying models for real-time prediction and monitoring necessitates efficient algorithms and hardware capable of quickly processing large volumes of data.	[53], [156]
----------------------	--	-------------

3) GENERALIZATION CHALLENGES

This study found that DL models for FF prediction often face significant challenges in generalizing when applied to new or unseen data. Overfitting was identified as a primary issue in which models performed well on the training data but struggled to generalize to new data [30], particularly when the training data were limited or not representative [150]. Additionally, the study highlighted that a shift in data distribution over time due to changing environmental conditions further impacted the model performance [58], [150].

Several strategies have been identified to address the generalization challenges faced by DL models for FF prediction. These include transfer learning, which leverages pre-trained models to enhance performance on new tasks, particularly when labeled data are limited [157], [158], and data augmentation, which involves artificially increasing the size and diversity of the training dataset through transformations, such as rotation and scaling of images, which can help models generalize better [158]. For example, data augmentation can create a more diverse training set from satellite images, thereby enhancing the generalizability of the model to new images [157], [158]. In addition, cross-validation techniques, which ensure that the model is validated on multiple data subsets, provide a more accurate estimate of its generalization performance [107]. Ensemble methods that combine predictions from multiple models can improve robustness and generalization [54], [63].

4) INTERPRETABILITY AND EXPLAINABILITY ISSUES

This study highlights the significant challenges in the interpretability and explainability of DL models for FF prediction. Despite their effectiveness, DL models for FF are often criticized for their "black box" nature, meaning that their decision-making processes are not transparent or easily understood [20], [56]. Interpretability and explainability are critical aspects of deep DL, particularly in applications such as FF prediction, in which understanding the decision-making process is essential for validation, trust, and actionable insights [22], [23]. Interpretability refers to the degree to which a human can understand the cause of a decision by using a model. Simultaneously, explainability pertains to how well the internal mechanics of a machine-learning system can be explained in human terms [33], [36].

The black-box nature of DL models, particularly deep neural networks (DNNs), makes it difficult to understand how decisions are made [87]. The complexity of these models can obscure the features that are most influential in the

predictions [78], [159]. Several strategies can be employed to address these challenges. Model simplification, although potentially at the cost of performance, can enhance the interpretability.

Several strategies have been developed to address this issue. These include model simplification [160], which involves reducing the complexity of the model to make it more understandable, and the use of SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) [105]. These techniques help quantify the importance of different features in the model's predictions, providing insights into which variables most influence the outcomes [160]. Additionally, visualization tools such as saliency maps have been noted for their ability to highlight parts of the input data that are most relevant to the model's predictions, offering a visual explanation that is easier to interpret [105].

VI. DISCUSSION

A. THEMATIC ANALYSIS OF EMERGING THEMES IN DL AND FF PREDICTION

Thematic analysis of emerging themes in DL and FF prediction reveals a strong emphasis on key concepts such as “deep learning,” “machine learning,” “artificial intelligence,” “neural networks,” and “remote sensing,” remote sensing, indicating that these technologies are central to the current research landscape. The inclusion of keywords like “Google Earth Engine,” “wildfires,” and “Sentinel-2” further underlines the reliance on satellite imagery and advanced computational tools for FF prediction. This thematic distribution shows the field's interdisciplinary nature, where AI, environmental science, and remote sensing converge to address the complex challenges in predicting and managing FFs.

A key challenge revealed by this thematic analysis was the need for better integration across interdisciplinary fields. Despite significant progress, the complexity of combining AI models with diverse data sources such as satellite imagery and environmental sensors remains a challenge that requires ongoing research and collaboration [17], [53].

Future research should focus on developing more seamless integration techniques to handle the complexity and variability of the data involved in FF prediction. Additionally, standardized methodologies are needed to bridge the gap between different disciplines, ensuring that the full potential of DL and AI in FF prediction is realized.

B. GLOBAL LANDSCAPE OF DL RESEARCH FOR FF PREDICTION

The global landscape of DL research for FF prediction shows a concentration of studies in countries such as the United States, China, France, Italy, India, Portugal, Greece, and Canada. These countries have made significant advances in this field, supported by their access to advanced computational resources, substantial investments in AI research, and

availability of diverse ecological data [71], [79]. This has led to the highest number of documents and citations, reflecting its leadership in AI research and technology.

This study also highlighted a significant disparity in adopting and developing DL technologies for FF prediction in Africa and other developing regions. Several studies have shown that the limited technological infrastructure in these areas causes this disparity [108], [161]. In addition, the scarcity of localized datasets and a gap in AI expertise have hindered their ability to keep pace with advancements in developed countries [56], [161]. This lack of representation poses a significant challenge because FF prediction models developed in different geographical and ecological contexts may not directly apply to regions with different environmental conditions.

To address this disparity, increasing investments in technological infrastructure, data collection, and AI expertise in developing regions are required. International collaborations and partnerships could be crucial in sharing knowledge, resources, and data, helping bridge the gap, and ensuring that DL models are applicable across different geographical and environmental contexts. Moreover, efforts should be made to develop localized datasets and models that can cater to these regions' specific needs and conditions, thus improving the global effectiveness of the FF prediction.

C. TRENDS OF PUBLICATION ON DL FOR FF PREDICTION OVER THE YEARS

The publication trends of DL for FF prediction from 2017 to 2024 revealed a significant surge in research activity between 2021 and 2022, with 2021 accounting for 36.36% of publications and 2022 peaking at 42.42%. This period of heightened activity can be attributed to several key factors, including the maturation of DL models, the increased availability of large and diverse datasets, and advancements in computational resources that facilitate more sophisticated research [70], [150]. In addition, the growing global awareness of the escalating frequency and severity of FFs has spurred interest and funding in this critical area [68].

However, the noticeable decline in research output observed in 2023 and 2024 indicates that several challenges still need to be solved despite progress. Several studies have shown that the primary issue is data heterogeneity, in which variations in data formats, resolutions, and collection methodologies from different sources create significant obstacles to the development of accurate and reliable prediction models [32], [65], [108]. In addition, diverse data sources and collection methodologies contribute to inconsistencies, complicating the development of accurate prediction models [56], [151]. This review found that, although most datasets are publicly accessible, their heterogeneity and varying standards present obstacles that necessitate rigorous preprocessing and standardization efforts.

The decline in publications also raises concerns about the sustainability of progress achieved during peak years because

the developed models do not have any sustainability plans [101], [119].

These challenges highlight the need for the research community to prioritize the development of standardized data collection and preprocessing protocols to address these issues. Moreover, there is a persistent need for improved data integration methods to harmonize datasets from various sources, ensuring that DL models are applicable across different FF prediction scenarios.

D. TYPE OF MODEL USED

Moreover, the analysis of the type of model used in FF prediction revealed the predominant use of classification models employed in 56% of the studies, emphasizing their effectiveness in categorizing data into predefined classes, such as 'fire' or 'no fire' [10], [69]. In addition, the regression models used in 24% of the studies are essential for predicting continuous outcomes such as fire size, duration, and spread rate, which are critical for understanding the dynamics of FFs [49], [88]. Although less common, object detection models are vital for identifying fire locations and boundaries in satellite imagery, highlighting their importance in specialized applications [10], [78]. Some studies have combined classification and regression approaches to demonstrate the value of integrating these techniques to address complex research questions.

However, the emphasis on classification and regression models suggests a potential gap in leveraging more advanced or hybrid modeling approaches [55]. Although these traditional models are compelling, several studies have shown the evolving complexity of FF prediction, which requires the exploration of new methodologies that can integrate multiple modeling techniques [17], [58]. The limited use of hybrid models that combine classification, regression, and object-detection techniques indicates an area ripe for innovation [79], [115].

This reliance on traditional models may restrict the potential of more comprehensive predictive frameworks, and several studies have highlighted the need for continued research on sophisticated models that can better handle the complexity and variability in FF predictions [79], [115]. Future research should prioritize developing and implementing advanced hybrid models to improve their ability to predict and manage FFs in increasingly complex scenarios.

E. TYPE OF STUDY CONDUCTED

Additionally, the study found a reliance on simulations compared with experiments. Simulation-based validation plays a significant role in developing and testing FF prediction models [53], [106]. Although simulations offer a controlled environment in which various scenarios can be efficiently tested, they have limitations [49], [52], [88]. For example, they rely on simplified models of reality and have the potential to overfit the simulated data [111]. Moreover, simulations may not accurately reflect the interplay between multiple environmental factors and human activities that influence fire behavior [55].

Several studies show that reliance on simulations raises concerns about the generalizability of the findings [27], [68], because the controlled nature of simulations does not capture the full spectrum of variability present in real-world scenarios, such as the impact of unexpected weather changes or unmodeled human activities [30], [68]. This limitation affects the model's applicability in diverse geographical and environmental contexts. Therefore, the generalizability of these findings is limited because models trained and validated in simulated environments may perform poorly when applied to real-world data [62], [150].

This expanded discussion highlights the representativeness of the datasets used in these studies. Many simulation-based studies have relied on datasets that may not fully represent the range of conditions in which FFs occur [95], [114]. For instance, simulations do not adequately model rare but critical events such as extreme weather conditions or large-scale human evacuation, leading to potential biases in model predictions [30].

The implications of these limitations are significant for the effectiveness of DL models in predicting FF. Without real-world validation, there is a risk that the models may be overfitted to the specific conditions of the simulations, thereby reducing their accuracy and reliability in practice [107]. Simulations cannot entirely replace insights gained from real-world experiments [55], [95]. This underscores the critical need for additional empirical studies to test these models in actual fire scenarios, thereby improving the strength and applicability of DL models for real-world FF predictions and management [68], [107]. The relatively low percentage of experimental studies highlights the practical challenges and constraints of conducting empirical research, such as high cost, logistical difficulties, and safety concerns [106], [111].

Incorporating real-world experiments is necessary to validate and enhance the reliability of FF prediction models [106]. Empirical testing provides critical insights into the model performance under actual conditions, revealing potential weaknesses and areas for improvement that simulations may overlook.

F. DL MODELS USED FOR FF PREDICTION

This study demonstrates that CNNs and LSTMs are the most commonly used DL models for FF prediction. CNNs and LSTMs were used in 48.78% and 12.20% of the studies, respectively. CNNs excel at processing spatial data such as satellite imagery [53], whereas LSTMs are effective for analyzing temporal data such as historical fire patterns [68], [71]. The rise of hybrid models, such as FU-NetCastV2, which combines CNNs and LSTMs, indicates a growing trend toward integrating multiple architectures to enhance predictive performance by capturing spatial and temporal data dependencies.

In addition to the use of traditional DL models like CNNs, LSTMs, and ANNs, remote sensing and environmental science contribute crucial datasets such as satellite

imagery, vegetation indices, and climate patterns [10], [27]. These datasets enhance the performance of DL models by enabling them to capture key environmental variables that influence fire behavior [13], [111]. Furthermore, incorporating human-induced factors, like land-use changes and urbanization, improves the ability of DL models to predict fire risks more accurately [101], [150]. These factors add socio-environmental dimensions to the models, making them better suited for understanding how human activities impact fire spread and ignition. This strengthens the adaptability of models in varied environmental contexts, ensuring better generalization and more reliable FF predictions.

Moreover, the diversity of models, with many appearing only once in the reviewed studies, highlights the experimental nature of research in this domain. Models such as DS-AttU, DS-Unet, DS-Unet++, WLF-Unet, Ensemble learning, All-ConvNet, and Residual U-Net indicate a broad exploration phase, in which researchers experimented with various architectures to determine the most effective approaches for specific types of data and prediction scenarios [59], [78]. This experimentation is essential for innovation but also suggests that the field is still in a developmental stage, with no single model emerging as the standard.

Advanced techniques, such as Deep CNNs (Fire-Net), BiLSTM+Attention, and ensemble learning, are also gaining popularity, reflecting a move towards more sophisticated models aiming to capture complex data patterns more effectively. These advanced models are crucial for improving prediction accuracy, particularly in handling the intricate interactions between various environmental factors that influence fire behavior [53], [58].

However, the diversity of models and the relatively equal distribution of different architectures present a significant challenge: the lack of standardization in the field. This variability makes it difficult to compare results across studies and hinders the development of universally applicable FF prediction models [108], [142]. The experimental nature of many models underscores the need to establish clear benchmarks and standardized evaluation criteria [108], [142]. Without these factors, it is challenging to assess the relative effectiveness of different models, which hampers the generalizability of the findings across different geographic and environmental contexts.

To address these challenges, future research should prioritize comparative studies that systematically evaluate the performance of various DL models under standardized conditions. This approach would help identify the most robust models and facilitate the development of more consistent and reliable prediction tools. Additionally, efforts should be made to align methodologies across studies, enabling researchers to build on each other's work more effectively and accelerating the advancement of the field.

Furthermore, there is a need to focus on hybrid models that have shown promise in combining the strengths of different architectures. Exploring these models in greater depth could lead to developing more comprehensive and accurate

frameworks for predicting the FF. As the field progresses, integrating standardized protocols and comparative benchmarks will be crucial for ensuring the practical applicability of DL models in real-world FF prediction and management scenarios.

G. DATASETS AND FEATURES

The analysis of mostly used datasets and relevant features for FF prediction provides important insights into the current state of research in this field. A review of 55 papers highlighted the strong reliance on imagery data such as satellite images, which underscores the importance of remote sensing technologies in monitoring and predicting FFs. This trend is complemented by the frequent integration of comprehensive climate and weather data, including critical variables, such as temperature, humidity, and wind speed, which are fundamental for understanding and predicting fire behavior [13], [111].

The emphasis on these traditional meteorological factors, temperature, humidity, and wind speed, as the top features used in FF prediction models, is expected, given their direct influence on fire dynamics. These factors are essential inputs for DL models aimed at predicting FF risk and behavior, as they directly affect vegetation dryness, ignition likelihood, and fire spread. Additionally, environmental features such as NDVI, rainfall, and other moisture-related indices were prominent, highlighting the importance of vegetation health and moisture content in assessing fire risk.

However, despite the comprehensive use of natural data, a significant gap remains in including human activity-related factors [101], [150]. This review revealed that factors such as proximity to roads, human settlements, agricultural activities, and population density were underrepresented in the reviewed datasets and models. This oversight is particularly concerning because human activities, including land-use changes, agricultural practices, and urbanization, play a critical role in fire ignition and spread [10], [27].

Including data on these factors, models can better capture the conditions under which fires are likely to ignite and spread. For instance, proximity to roads and population density can indicate areas of higher fire risk due to increased human interaction with forested areas. Additionally, incorporating data on activities like logging, mining, or recreational use of forest lands can help models predict fire outbreaks more accurately. Underrepresentation of these anthropogenic factors leads to an incomplete understanding of fire risk, thereby limiting the effectiveness of FF prediction models in regions where human influence is significant [151], [162].

Moreover, analyzing the datasets used in these studies revealed several challenges related to data quality and consistency. Although the majority of datasets are publicly available, variability in data formats, resolutions, and collection methodologies from different sources poses significant obstacles to the development of accurate and reliable prediction models [56], [108]. This data heterogeneity makes seamless integration difficult and necessitates extensive pre-processing and standardization [37], [141].

Future research should prioritize the inclusion of human activity data in FF prediction models to capture the complex interplay between natural and human factors in fire dynamics. In addition, the development of standardized data collection and preprocessing protocols is critical to improve the reliability and applicability of these models across different geographical and environmental contexts.

H. COMMONLY USED SATELLITE PLATFORMS FOR FF PREDICTION

This analysis highlights the critical role of satellite platforms in FF prediction, with Landsat being the most frequently used platform, accounting for 25% of studies. This is likely because of the extensive archive of high-resolution imagery, which is crucial for training DL models for historical fire events [113], [128]. MODIS and Sentinel satellites, each used in 13% of the studies, are favored for their global coverage and advanced imaging capabilities, making them effective for real-time fire detection and detailed spatial analysis, even with smoke and cloud cover [113], [128].

Other platforms such as VIIRS, GOES, and the NOAA Visible Infrared Imaging Radiometer Suite are valued for their infrared sensing capabilities, which are essential for detecting heat anomalies associated with fires [17], [27]. The inclusion of aerial photos and Google Earth Satellite Images in a smaller percentage of studies suggests their use in providing higher-resolution imagery to complement satellite data, particularly for validating fire detection models [108], [142].

Despite the advantages offered by these diverse satellite platforms, this study revealed significant challenges in integrating data from multiple sources [108], [142], as a single study may use satellite data or images from multiple platforms. Differences in resolution, spectral capabilities, and temporal coverage across these platforms pose substantial obstacles to the creation of consistent and accurate FF prediction models [154]. Reliance on multiple platforms necessitates extensive preprocessing and standardization efforts to harmonize the data, ensuring that predictions are accurate and reliable across various geographical contexts.

This complexity in data integration emphasizes the need to develop standardized methodologies that effectively combine data from different satellite sources. Such standards are necessary for potential discrepancies and inaccuracies in prediction models to remain high, which can undermine the effectiveness of FF management strategies. Future research should prioritize the creation of data integration frameworks that can leverage the unique strengths of each satellite platform, ensuring that DL models for FF prediction are comprehensive and reliable.

I. PERFORMANCE EVALUATION METRICS

The analysis of performance evaluation metrics in FF prediction models revealed a strong reliance on traditional metrics such as accuracy, precision, recall, F1 score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

These metrics are fundamental in assessing the effectiveness of DL (DL) models by providing insights into how well these models can predict FFs while minimizing false positives and negatives. The frequent use of these metrics, as summarized in Figure 18, underscores their importance in ensuring that the models are reliable and effective in their predictive capabilities.

However, recent studies have highlighted the limitations of relying solely on traditional metrics. Although they are useful for gauging the overall model performance, they may not fully capture the complexities of the FF prediction, particularly in real-world scenarios [110], [139]. For instance, these metrics may not account for the spatial and temporal dynamics of fires, which are crucial for understanding and predicting fire behavior across different regions and times [142]. This can be especially problematic in regions where fires are infrequent but potentially devastating when they do occur. In such cases, models might perform well according to standard metrics but fail to provide accurate predictions when they are most needed [90], [142].

Moreover, current evaluation metrics may not adequately reflect the challenges of different geographic regions, where environmental and climatic conditions vary significantly [128], [139]. This oversight could lead to models that perform well in controlled environments or simulations but struggle with the complexities of real-world applications.

To address these challenges, more comprehensive evaluation frameworks that go beyond the traditional metrics are required. Future research should focus on developing these frameworks and incorporating context-specific performance indicators that consider the unique challenges of FF predictions in various environments.

J. DATA LIMITATIONS

Upon analyzing the data limitations, the primary challenge identified in this study is the availability and accessibility of high-quality data. Recent studies have also shown that obtaining high-resolution relevant data is particularly difficult in high-risk areas where FFs are the most prevalent [56], [57]. Furthermore, historical data often fail to capture current dynamics, such as recent land-use changes or shifts in fire susceptibility due to climate change [120], [129]. This limitation in data availability can lead to outdated or less effective models for predicting contemporary fire behavior.

Another significant challenge is the volume and diversity of the data. Handling large datasets, including various data types, such as vegetation indices, climatic conditions, and topography, requires sophisticated spatial analysis and advanced modeling techniques [65], [150]. The complexity of these datasets necessitates extensive preprocessing, which can be resource-intensive and technically demanding.

This study also highlighted the need for real-time data integration. Real-time data are crucial for predicting and responding to FFs as conditions evolve. However, integrating such data is challenging owing to logistical and technical constraints [108], [142]. Historical datasets do not fully capture

the rapidly changing conditions that can lead to FFs, thereby reducing the effectiveness of models when applied in real-time scenarios.

Furthermore, integration of human activity data remains a significant challenge. Human activities such as land-use changes, agricultural practices, and urbanization are critical factors in fire ignition and spread. However, these data are more complex to collect and standardize than environmental data, leading to underrepresentation in the FF prediction models [141], [162]. Incorporating human activity data into models increases their complexity and requires sophisticated approaches and resources for development and analysis [141].

Advanced and integrated data-collection methods are critical for overcoming these challenges. Improved data collection infrastructure and methodologies are essential for acquiring high-quality, high-resolution data and developing robust and reliable FF prediction models.

K. CHALLENGES IN EVALUATING DL MODELS FOR FF PREDICTION

Evaluating DL models for FF prediction presents challenges, particularly related to data imbalance, temporal and spatial variability, and the need for extensive data preprocessing.

Data imbalance is a pervasive issue in the prediction of FF [10]. In many datasets, non-fire events significantly outnumber actual fire occurrences, leading to models that may overestimate accuracy while failing to detect rare but critical fire events [111], [113]. This imbalance can result in less effective models for real-world applications, where accurate prediction of rare events is crucial [108].

The temporal and spatial variability of FFs also pose significant challenges to model generalizability. Fires are influenced by seasonal changes, weather conditions, and geographical diversity, making it necessary to test models under various conditions to ensure their reliability in different environments [150]. Models that perform well in one region or under certain conditions may not be generalizable to other scenarios, thus highlighting the need for more comprehensive testing and validation [142].

Data preprocessing and interpretability are critical challenges. The complex task of preparing data for model training, combined with the difficulty in interpreting DL models, particularly in the context of FF prediction, limits their practical application. Understanding how models make predictions is crucial for their adoption in real-world settings where trust and validation are essential [56], [163].

Moreover, the complexity of the FF phenomena influenced by various factors, such as vegetation type, meteorological conditions, topography, and human activities, makes accurate modeling a significant challenge [142]. The evolving nature of technology and data collection methods adds another layer of complexity that requires models to be continually updated and evaluated against new data and methodologies [163].

Finally, optimizing DL models, including hyperparameter tuning and the need for significant computational resources,

presents additional barriers [53]. The time-consuming and complex nature of optimizing models to achieve the best performance can limit their accessibility and applicability, particularly in resource-constrained settings [67].

To address these challenges, future research should develop innovative data augmentation techniques, sophisticated model optimization methods, and efficient algorithms that reduce computational demands.

L. GENERALIZATION CHALLENGES

The results emphasize a critical challenge in deploying DL models for FF prediction owing to the difficulty in generalizing to new or unseen data. This is particularly concerning in the context of FF prediction, where models must accurately predict fire risk across different regions under various conditions [147]. Overfitting, in which models perform well on training data but fail with new data, highlights the need for more robust training datasets that are representative of the full range of conditions under which FFs occur [109], [114].

The impact of data distribution shifts due to changing environmental conditions further complicates the generalization of these models. As environmental conditions evolve, models trained using past data may become less accurate, thereby underscoring the need for models that can adapt to such changes.

Future research should focus on refining and advancing the identified strategies to address these challenges. Transfer learning and data augmentation offer promising avenues for improving the model generalization [27], [29]. However, there is a need to explore further how these techniques can be optimized specifically for FF prediction. Ensemble methods and robust cross-validation protocols should be developed further to enhance the generalizability and robustness of DL models in this critical application area [33], [120]. Moreover, integrating these strategies into real-time FF prediction systems ensures that the models adapt to changing conditions and remain effective in diverse, dynamic environments.

M. INTERPRETABILITY AND EXPLAINABILITY ISSUES

The findings underscore a significant limitation in using DL models for FF prediction of their ‘black box’ nature. This opacity not only makes it challenging to validate the models but also raises concerns about their reliability in real-world scenarios [78], [132]. The interpretability of DL models is crucial because understanding how a model arrives at its decisions can inform more effective fire management strategies. Although techniques such as SHAP, LIME, and saliency maps offer promising avenues for enhancing interpretability [105], these approaches often involve trade-offs such as potentially reduced model performance [160]. This balance between maintaining high predictive accuracy and ensuring interpretability is a critical area that requires further exploration.

To overcome these challenges, future research should focus on refining methods to enhance interpretability without significantly compromising the performance of the DL

models. Developing standardized frameworks for assessing and improving the transparency of these models is essential. Moreover, integrating these refined techniques into FF prediction models can improve stakeholder trust and facilitate practical application in real-world FF management systems. As the field advances, it is important to explore how these interpretability tools can be scaled and adapted to handle the increasing complexity of the data and model architectures for FF predictions.

VII. CONCLUSION

This systematic review illustrates the transformative potential of DL in FF prediction, showcasing its interdisciplinary application in remote sensing, environmental science, and artificial intelligence. The analysis strongly emphasizes CNNs and LSTMs as the most frequently used models, with CNNs excelling in spatial data processing and LSTMs capturing temporal dependencies. These models, alongside emerging hybrid architectures like FU-NetCastV2, indicate a progressive trend toward integrating multiple techniques to enhance prediction accuracy.

Key findings highlight the central role of meteorological variables such as temperature, humidity, and wind speed, complemented by satellite-derived indices like NDVI, LST, and NDMI, in improving model reliability. The analysis of datasets emphasizes a significant reliance on remote sensing data, particularly from platforms such as Landsat, MODIS, and Sentinel, which provide critical inputs for FF prediction models. However, the underrepresentation of human activity data, including land-use changes and proximity to roads, emerges as a critical gap, limiting the holistic applicability of these models in regions where anthropogenic factors heavily influence fire dynamics.

The review identifies global disparities in research output, with developed nations like the United States, China, and France leading advancements in DL for FF prediction. These countries benefit from superior computational resources and diverse datasets, enabling robust modeling efforts. In contrast, developing regions, including Africa, face challenges such as limited technological infrastructure and a lack of localized datasets, underscoring the urgent need for investments and collaborations to bridge this gap.

Trends in publication activity reflect the field's growth, particularly during 2021-2023, driven by advancements in DL and increased computational resources. However, a decline in research output in 2024 highlights unresolved challenges such as data heterogeneity, variability in data quality and formats, and the lack of standardized protocols for data integration. These issues hinder the comparability of studies and the scalability of DL models for diverse geographical contexts.

The study also exposes a heavy reliance on simulations (68% of reviewed studies) over real-world experiments, raising concerns about models' generalizability and practical applicability. While valuable for initial validation, simulations cannot replicate the complex interplay of environmental

and anthropogenic factors in real-world scenarios. Consequently, integrating field-based testing into future research is essential to strengthen model validation and applicability.

The analysis of performance evaluation metrics reveals a reliance on traditional measures such as accuracy, precision, recall, and F1 score, which, while useful, may not fully capture the complexities of FF dynamics. The need for context-specific evaluation frameworks considering spatial, temporal, and regional variability is evident. Additionally, data imbalance, preprocessing, and model interpretability challenges highlight the importance of advancing techniques such as SHAP and saliency maps to improve transparency and trust in DL models.

A. LIMITATIONS OF THE STUDY

This review is limited by its focus on specific DL methodologies, primarily covering studies from developed regions. The timeframe of reviewed publications may not capture the latest advancements, and the exclusion of unconventional approaches may overlook innovative trends. Expanding future reviews to encompass a broader range of methodologies, datasets, and geographic contexts will provide a more comprehensive understanding of the field.

B. FUTURE RESEARCH

Future research should develop datasets incorporating human activity metrics, establish standardized data protocols, and emphasize real-world validation through field testing. Collaboration among technologists, ecologists, and communities is vital for inclusive solutions. Advanced evaluation frameworks addressing regional variability and socioeconomic factors are needed for standardized benchmarking. Additionally, exploring emerging AI techniques, such as unsupervised learning, reinforcement learning, and hybrid approaches, can enhance model robustness, generalizability, and effectiveness in FF prediction and management.

REFERENCES

- [1] R. Fischer, "The long-term consequences of forest fires on the carbon fluxes of a tropical forest in Africa," *Appl. Sci.*, vol. 11, no. 10, p. 4696, May 2021, doi: [10.3390/app11104696](https://doi.org/10.3390/app11104696).
- [2] P. Joseph Kagosi, S. E. Nkya, C. P. Balama, and I. H. Babili, "Effectiveness of firewise approach (FWA) in controlling forest fire: A case of Uluguru Nature Forest Reserve (UNFR), Morogoro, Tanzania," *J. Agricult. Sci. Technol. B*, vol. 10, no. 3, pp. 175–185, Jun. 2020, doi: [10.17265/2161-6264/2020.03.006](https://doi.org/10.17265/2161-6264/2020.03.006).
- [3] R. Pillay, M. Venter, J. Aragon-Osejo, P. González-del-Piego, A. J. Hansen, J. E. Watson, and O. Venter, "Tropical forests are home to over half of the world's vertebrate species," *Frontiers Ecol. Environ.*, vol. 20, no. 1, pp. 10–15, Feb. 2022, doi: [10.1002/fee.2420](https://doi.org/10.1002/fee.2420).
- [4] N. L. Michel, C. J. Whelan, and G. M. Verutes, "Ecosystem services provided by neotropical birds," *Condor*, vol. 122, no. 3, Sep. 2020, Art. no. duaa022, doi: [10.1093/condor/duaa022](https://doi.org/10.1093/condor/duaa022).
- [5] G. Bayat and K. Yildiz, "Predicting wildfires using machine learning methods," *Turkish J. Sci. Technol.*, vol. 17, no. 1, pp. 241–250, Apr. 2022, doi: [10.55525/tjst.1063284](https://doi.org/10.55525/tjst.1063284).
- [6] Y. O. Sayad, H. Mousannif, and H. Al Moatassime, "Predictive modeling of wildfires: A new dataset and machine learning approach," *Fire Saf. J.*, vol. 104, pp. 130–146, Mar. 2019, doi: [10.1016/j.firesaf.2019.01.006](https://doi.org/10.1016/j.firesaf.2019.01.006).
- [7] H. Stipp, *US Displacements By Wildfires 2008-2022* | Statista. Accessed: Jul. 20, 2024. [Online]. Available: <https://www.statista.com/statistics/1422376/number-people-displaced-by-wildfires-usa/>

- [8] IF10244, Congressional Res. Service, Washington, DC, USA, 2023.
- [9] B. E. Salas. *Global Deaths Due To Wildfires 2023* | Statista. Accessed: Jul. 21, 2024. [Online]. Available: <https://www.statista.com/statistics/1293254/global-number-of-deaths-due-to-wildfires/>
- [10] P. Dr. Urmila Shrawankar, H. Pazare, P. Raut, M. Dhanorkar, and S. Dhage, "Forest fire prediction using deep learning," *Int. J. Adv. Res. Sci., Commun. Technol.*, vol. 2, no. 1, pp. 465–470, Dec. 2022, doi: 10.48175/ijarsct-7711.
- [11] D. M. J. S. Bowman, C. A. Kolden, J. T. Abatzoglou, F. H. Johnston, G. R. van der Werf, and M. Flannigan, "Vegetation fires in the anthropocene," *Nature Rev. Earth Environ.*, vol. 1, no. 10, pp. 500–515, Aug. 2020, doi: 10.1038/s43017-020-0085-3.
- [12] P. Deb, H. Moradkhani, P. Abbaszadeh, A. S. Kiem, J. Engström, D. Keellings, and A. Sharma, "Causes of the widespread 2019–2020 Australian bushfire season," *Earth's Future*, vol. 8, no. 11, Nov. 2020, Art. no. e2020EF001671, doi: 10.1029/2020ef001671.
- [13] H. Liang, M. Zhang, and H. Wang, "A neural network model for wildfire scale prediction using meteorological factors," *IEEE Access*, vol. 7, pp. 176746–176755, 2019, doi: 10.1109/ACCESS.2019.2957837.
- [14] N. J. Abram, B. J. Henley, A. Sen Gupta, T. J. R. Lippmann, H. Clarke, A. J. Dowdy, J. J. Sharples, R. H. Nolan, T. Zhang, M. J. Wooster, J. B. Wurtzel, K. J. Meissner, A. J. Pitman, A. M. Ukkola, B. P. Murphy, N. J. Tapper, and M. M. Boer, "Connections of climate change and variability to large and extreme forest fires in Southeast Australia," *Commun. Earth Environ.*, vol. 2, no. 1, p. 8, Jan. 2021, doi: 10.1038/s43247-020-00065-8.
- [15] A. M. S. Copernicus. 2023: *A Year of Intense Global Wildfire Activity* | Copernicus. Accessed: Jul. 21, 2024. [Online]. Available: <https://atmosphere.copernicus.eu/2023-year-intense-global-wildfire-activity>
- [16] F. Abid, "A survey of machine learning algorithms based forest fires prediction and detection systems," Mar. 01, 2021, Springer, doi: 10.1007/s10694-020-01056-z.
- [17] I. Prapas, A. Ahuja, S. Kondylatos, I. Karasante, E. Panagiotou, L. Alonso, C. Davalas, D. Michail, N. Carvalhais, and I. Papoutsis, "Deep learning for global wildfire forecasting," 2022, *arXiv:2211.00534*.
- [18] S. Galal. *Africa: Annual Forest Area Net Change* | Statista. Accessed: Jul. 21, 2024. [Online]. Available: <https://www.statista.com/statistics/1286821/annual-forest-area-net-change-in-africa/>
- [19] M. Mgina and A. Wawa, "Assessment of wildfires in Tanzania forest plantations: A case of Sao Hill in Mufindi District," *Huria, J. Open Univ. Tanzania*, vol. 27, no. 2, pp. 104–112, Sep. 2020.
- [20] M. Grari, M. Yandouzi, I. Idrissi, M. Boukabous, O. Moussaoui, M. Azizi, and M. Moussaoui, "Using IoT and ML for forest fire detection, monitoring, and prediction: A literature review," *J. Theor. Appl. Inf. Technol.*, vol. 15, p. 19, 2022. [Online]. Available: www.jatit.org
- [21] I. El-Madafri, M. Peña, and N. Olmedo-Torre, "The wildfire dataset: Enhancing deep learning-based forest fire detection with a diverse evolving open-source dataset focused on data representativeness and a novel multi-task learning approach," *Forests*, vol. 14, no. 9, p. 1697, Aug. 2023, doi: 10.3390/f14091697.
- [22] G. Zhang, M. Wang, and K. Liu, "Forest fire susceptibility modeling using a convolutional neural network for Yunnan province of China," *Int. J. Disaster Risk Sci.*, vol. 10, no. 3, pp. 386–403, Sep. 2019, doi: 10.1007/s13753-019-00233-1.
- [23] M. Azami, N. Orger, V. Schulz, T. Oshiro, and M. Cho, "Earth observation mission of a 6U CubeSat with a 5-Meter resolution for wildfire image classification using convolution neural network approach," *Remote Sens.*, vol. 14, no. 8, p. 1874, Apr. 2022, doi: 10.3390/rs14081874.
- [24] M. H. Shams Eddin, R. Roscher, and J. Gall, "Location-aware adaptive normalization: A deep learning approach for wildfire danger forecasting," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Dec. 2023, Art. no. 4703018, doi: 10.1109/TGRS.2023.3285401.
- [25] C. Ding, X. Zhang, J. Chen, S. Ma, Y. Lu, and W. Han, "Wildfire detection through deep learning based on Himawari-8 satellites platform," *Int. J. Remote Sens.*, vol. 43, no. 13, pp. 5040–5058, Jul. 2022, doi: 10.1080/01431161.2022.2119110.
- [26] M. A. Jamshed, C. Theodorou, T. Kalsoom, N. Anjum, Q. H. Abbasi, and M. Ur-Rehman, "Intelligent computing based forecasting of deforestation using fire alerts: A deep learning approach," *Phys. Commun.*, vol. 55, Dec. 2022, Art. no. 101941, doi: 10.1016/j.phycom.2022.101941.
- [27] S. Kondylatos, I. Prapas, M. Ronco, I. Papoutsis, G. Camps-Valls, M. Piles, M. Fernández-Torres, and N. Carvalhais, "Wildfire danger prediction and understanding with deep learning," *Geophys. Res. Lett.*, vol. 49, no. 17, pp. 1–11, Sep. 2022, doi: 10.1029/2022gl099368.
- [28] A. M. Prajith and A. Prashant, "Forest wildfire detection from satellite images using deep learning," *Int. J. Adv. Res. Sci., Commun. Technol.*, vol. 3, no. 1, pp. 578–583, Jul. 2023, doi: 10.48175/ijarsct-12086.
- [29] M. Naderpour, H. M. Rizeei, and F. Ramezani, "Forest fire risk prediction: A spatial deep neural network-based framework," *Remote Sens.*, vol. 13, no. 13, p. 2513, Jun. 2021, doi: 10.3390/rs13132513.
- [30] J. L. Hodges and B. Y. Lattimer, "Wildland fire spread modeling using convolutional neural networks," *Fire Technol.*, vol. 55, no. 6, pp. 2115–2142, Nov. 2019, doi: 10.1007/s10694-019-00846-4.
- [31] M. Elliot, "What is AI?" in *Architecture in the Age of Artificial Intelligence*, N. Leach, Ed. London, U.K.: Bloomsbury Publishing Plc, 2021, pp. 15–40, doi: 10.5040/9781350165557.ch-001.
- [32] U. Mittal and D. M. Sharma, "Artificial intelligence and its application in different areas of Indian economy," *Int. J. Adv. Res. Sci., Commun. Technol.*, vol. 2, no. 3, pp. 160–163, Feb. 2021, doi: 10.48175/IJARSCT-V2-I3-328.
- [33] P. K. Garg, "Overview of artificial intelligence," in *Artificial Intelligence*. Boca Raton, FL, USA: Chapman, 2021, pp. 3–18, doi: 10.1201/9781003140351-2.
- [34] M. W. Jones, J. T. Abatzoglou, S. Veraverbeke, N. Andela, G. Lasslop, M. Forkel, A. J. P. Smith, C. Burton, R. A. Betts, G. R. van der Werf, S. Sitch, J. G. Canadell, C. Santin, C. Kolden, S. H. Doerr, and C. Le Quéré, "Global and regional trends and drivers of fire under climate change," *Rev. Geophysics*, vol. 60, no. 3, Sep. 2022, Art. no. e2020RG000726, doi: 10.1029/2020rg000726.
- [35] Y. Jiang, X. Li, H. Luo, S. Yin, and O. Kaynak, "Quo vadis artificial intelligence?" *Discover Artif. Intell.*, vol. 2, no. 1, p. 4, Mar. 2022, doi: 10.1007/s44163-022-00022-8.
- [36] W. Wang, "Using machine learning," in *Pro iPhone Development With SwiftUI*. Berkeley, CA, USA: Apress, 2022, pp. 303–324, doi: 10.1007/978-1-4842-7827-7_17.
- [37] A. Mohapatra and T. Trinh, "Early wildfire detection technologies in practice—A review," *Sustainability*, vol. 14, no. 19, p. 12270, 2022, doi: 10.3390/su141912270.
- [38] M. Jutel, M. Zemelka-Wiacek, M. Ordak, O. Pfaar, T. Eiwegger, M. Rechenmacher, and C. A. Akdis, "The artificial intelligence (AI) revolution: How important for scientific work and its reliable sharing," *Allergy*, vol. 78, no. 8, pp. 2085–2088, Aug. 2023, doi: 10.1111/all.15778.
- [39] Y. Li, Z. Feng, S. Chen, Z. Zhao, and F. Wang, "Application of the artificial neural network and support vector machines in forest fire prediction in the Guangxi autonomous region, China," *Discrete Dyn. Nature Soc.*, vol. 2020, pp. 1–14, Apr. 2020, doi: 10.1155/2020/5612650.
- [40] L. K. Sharma, R. Gupta, and N. Fatima, "Assessing the predictive efficacy of six machine learning algorithms for the susceptibility of Indian forests to fire," *Int. J. Wildland Fire*, vol. 31, no. 8, pp. 735–758, Jul. 2022, doi: 10.1071/wf22016.
- [41] B. R. Ilyas, B. M. Sofiane, T. A. Abderrazak, K. Miloud, F. Kamila, and B. Boukenadil, "Forest fire detection with combined SVM and deep CNN approach," in *Proc. 2nd Int. Conf. Electr. Eng. Autom. Control (ICEEAC)*, May 2024, pp. 1–6, doi: 10.1109/iceeac61226.2024.10576417.
- [42] S. Yang, M. Lupascu, and K. S. Meel, "Predicting forest fire using remote sensing data and machine learning," in *Proc. 35th AAAI Conf. Artif. Intell.*, 2021, pp. 14983–14990, doi: 10.1609/aaai.v35i17.17758.
- [43] F. Huot, R. L. Hu, N. Goyal, T. Sankar, M. Ihme, and Y.-F. Chen, "Next day wildfire spread: A machine learning dataset to predict wildfire spreading from remote-sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4412513, doi: 10.1109/TGRS.2022.3192974.
- [44] D. Makowski, "Simple random forest classification algorithms for predicting occurrences and sizes of wildfires," *Extremes*, vol. 26, no. 2, pp. 331–338, Jun. 2023, doi: 10.1007/s10687-022-00458-2.
- [45] A. Bjånes, R. De La Fuente, and P. Mena, "A deep learning ensemble model for wildfire susceptibility mapping," *Ecological Informat.*, vol. 65, Nov. 2021, Art. no. 101397, doi: 10.1016/j.ecoinf.2021.101397.
- [46] S. Khanmohammadi, M. Arashpour, E. M. Golafrshani, M. G. Cruz, and A. Rajabifard, "An artificial intelligence framework for predicting fire spread sustainability in semiarid shrublands," *Int. J. Wildland Fire*, vol. 32, no. 4, pp. 636–649, Feb. 2023, doi: 10.1071/wf22216.
- [47] G. Chitrakleha and J. M. Roogi, "A quick review of ML algorithms," in *Proc. 6th Int. Conf. Commun. Electronics Syst. (ICCES)*, Jul. 2021, pp. 1–5, doi: 10.1109/ICCES51350.2021.9488982.
- [48] M. Rodrigues and J. de la Riva, "An insight into machine-learning algorithms to model human-caused wildfire occurrence," *Environ. Model. Softw.*, vol. 57, pp. 192–201, Jul. 2014, doi: 10.1016/j.envsoft.2014.03.003.

- [49] Y. Pang, Y. Li, Z. Feng, Z. Feng, Z. Zhao, S. Chen, and H. Zhang, "Forest fire occurrence prediction in China based on machine learning methods," *Remote Sens.*, vol. 14, no. 21, p. 5546, Nov. 2022, doi: [10.3390/rs14215546](https://doi.org/10.3390/rs14215546).
- [50] P. Karthikeyan, K. Velswamy, P. Harshavardhanan, R. Rajagopal, V. Jeyakrishnan, and S. Velliangiri, "Machine learning techniques application," in *Research Anthology on Architectures, Frameworks, and Integration Strategies for Distributed and Cloud Computing*, Hershey, PA, USA: IGI Global, 2021, pp. 1396–1417, doi: [10.4018/978-1-7998-5339-8.ch068](https://doi.org/10.4018/978-1-7998-5339-8.ch068).
- [51] A. Trucchia, H. Izadgoshasb, S. Isnardi, P. Fiorucci, and M. Tonini, "Machine-learning applications in geosciences: Comparison of different algorithms and vegetation classes' importance ranking in wildfire susceptibility," *Geosciences*, vol. 12, no. 11, p. 424, Nov. 2022, doi: [10.3390/geosciences12110424](https://doi.org/10.3390/geosciences12110424).
- [52] W. Benzekri, A. El, O. Moussaoui, and M. Berrajaa, "Early forest fire detection system using wireless sensor network and deep learning," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 5, pp. 496–503, 2020, doi: [10.14569/ijacsa.2020.0110564](https://doi.org/10.14569/ijacsa.2020.0110564).
- [53] M. Marjani, M. Mahdianpari, and F. Mohammadimanesh, "CNN-BiLSTM: A novel deep learning model for near-real-time daily wildfire spread prediction," *Remote Sens.*, vol. 16, no. 8, p. 1467, Apr. 2024.
- [54] O. Oak, R. Nazre, S. Naigaonkar, S. Sawant, and A. Joshi, "A novel transfer learning based CNN model for wildfire susceptibility prediction," in *Proc. 5th Int. Conf. Emerg. Technol. (INCET)*, May 2024, pp. 1–6, doi: [10.1109/incet61516.2024.10593496](https://doi.org/10.1109/incet61516.2024.10593496).
- [55] S. Monaco, S. Greco, A. Farasin, L. Colomba, D. Apiletti, P. Garza, T. Cerquitelli, and E. Baralis, "Attention to fires: Multi-channel deep learning models for wildfire severity prediction," *Appl. Sci.*, vol. 11, no. 22, p. 11060, Nov. 2021, doi: [10.3390/app112211060](https://doi.org/10.3390/app112211060).
- [56] R. Ghali and M. A. Akhloufi, "Deep learning approaches for wildland fires using satellite remote sensing data: Detection, mapping, and prediction," *Fire*, vol. 6, no. 5, p. 192, 2023, doi: [10.3390/fire6050192](https://doi.org/10.3390/fire6050192).
- [57] S. Zheng, P. Gao, W. Wang, and X. Zou, "A highly accurate forest fire prediction model based on an improved dynamic convolutional neural network," *Appl. Sci.*, vol. 12, no. 13, p. 6721, Jul. 2022, doi: [10.3390/app12136721](https://doi.org/10.3390/app12136721).
- [58] M. Marjani, S. A. Ahmadi, and M. Mahdianpari, "FirePred: A hybrid multi-temporal convolutional neural network model for wildfire spread prediction," *Ecol. Informat.*, vol. 78, Dec. 2023, Art. no. 102282, doi: [10.1016/j.ecoinf.2023.102282](https://doi.org/10.1016/j.ecoinf.2023.102282).
- [59] S. Natekar, S. Patil, A. Nair, and S. Roychowdhury, "Forest fire prediction using LSTM," in *Proc. 2nd Int. Conf. Emerg. Technol. (INCET)*, May 2021, pp. 1–5, doi: [10.1109/INCET51464.2021.9456113](https://doi.org/10.1109/INCET51464.2021.9456113).
- [60] J. Li, T. Li, R. Lin, and Q. Nie, "GAN-based models and applications," in *Proc. IEEE 5th Int. Conf. Inf. Syst. Comput. Aided Educ. (ICISCAE)*, Sep. 2022, pp. 848–852, doi: [10.1109/ICISCAE55891.2022.9927647](https://doi.org/10.1109/ICISCAE55891.2022.9927647).
- [61] R. Ganz, *A Review of Generative Adversarial Networks*. Accessed: Jul. 30, 2024. [Online]. Available: <https://medium.com/analytics-vidhya/a-review-of-generative-adversarial-networks-part-1-a3e5757a3dc2>
- [62] A. Santopaolo, S. S. Saif, A. Pietrabissa, and A. Giuseppe, "Forest fire risk prediction from satellite data with convolutional neural networks," in *Proc. 29th Medit. Conf. Control Autom. (MED)*, Jun. 2021, pp. 360–367, doi: [10.1109/MED51440.2021.9480226](https://doi.org/10.1109/MED51440.2021.9480226).
- [63] H. C. Reis and V. Turk, "Detection of forest fire using deep convolutional neural networks with transfer learning approach," *Appl. Soft Comput.*, vol. 143, Aug. 2023, Art. no. 110362, doi: [10.1016/j.asoc.2023.110362](https://doi.org/10.1016/j.asoc.2023.110362).
- [64] W. Jiang, Y. Qiao, G. Su, X. Li, Q. Meng, H. Wu, W. Quan, J. Wang, and F. Wang, "WFNet: A hierarchical convolutional neural network for wildfire spread prediction," *Environ. Model. Softw.*, vol. 170, Dec. 2023, Art. no. 105841, doi: [10.1016/j.envsoft.2023.105841](https://doi.org/10.1016/j.envsoft.2023.105841).
- [65] M. Marjani and M. S. Mesgari, "The large-scale wildfire spread prediction using a multi-kernel convolutional neural network," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 10, pp. 483–488, Jan. 2023, doi: [10.5194/isprs-annals-x-4-w1-2022-483-2023](https://doi.org/10.5194/isprs-annals-x-4-w1-2022-483-2023).
- [66] X. Li, X. Wang, S. Sun, Y. Wang, S. Li, and D. Li, "Predicting the wildland fire spread using a mixed-input CNN model with both channel and spatial attention mechanisms," *Fire Technol.*, vol. 59, no. 5, pp. 2683–2717, Sep. 2023, doi: [10.1007/s10694-023-01427-2](https://doi.org/10.1007/s10694-023-01427-2).
- [67] M. Marjani, M. Mahdianpari, S. A. Ahmadi, E. Hemmati, F. Mohammadimanesh, and M. S. Mesgari, "Application of explainable artificial intelligence in predicting wildfire spread: An ASPP-enabled CNN approach," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, doi: [10.1109/LGRS.2024.3417624](https://doi.org/10.1109/LGRS.2024.3417624).
- [68] R. Kanwal, W. Rafiqat, M. Iqbal, and S. Weiguo, "Data-driven approaches for wildfire mapping and prediction assessment using a convolutional neural network (CNN)," *Remote Sens.*, vol. 15, no. 21, p. 5099, Oct. 2023, doi: [10.3390/rs15215099](https://doi.org/10.3390/rs15215099).
- [69] Y. Kang, E. Jang, J. Im, and C. Kwon, "A deep learning model using geostationary satellite data for forest fire detection with reduced detection latency," *GIScience Remote Sens.*, vol. 59, no. 1, pp. 2019–2035, Dec. 2022, doi: [10.1080/15481603.2022.2143872](https://doi.org/10.1080/15481603.2022.2143872).
- [70] Q. Yu, Y. Zhao, Z. Yin, and Z. Xu, "Wildfire susceptibility prediction based on a CA-based CCNN with active learning optimization," *Fire*, vol. 7, no. 6, p. 201, Jun. 2024, doi: [10.3390/fire7060201](https://doi.org/10.3390/fire7060201).
- [71] G. Zhang, M. Wang, and K. Liu, "Deep neural networks for global wildfire susceptibility modelling," *Ecol. Indicators*, vol. 127, Aug. 2021, Art. no. 107735, doi: [10.1016/j.ecolind.2021.107735](https://doi.org/10.1016/j.ecolind.2021.107735).
- [72] W. S. Wolfgang, *Machine Learning for Brain Disorders*. Berlin, Germany: Springer, 2023. [Online]. Available: <http://www.springer.com/series/7657>
- [73] R. Perumal and T. L. V. Zyl, "Comparison of recurrent neural network architectures for wildfire spread modelling," in *Proc. Int. SAUPEC/RobMech/PRASA Conf.*, Jan. 2020, pp. 1–6, doi: [10.1109/SAUPEC/RobMech/PRASA48453.2020.9078028](https://doi.org/10.1109/SAUPEC/RobMech/PRASA48453.2020.9078028).
- [74] I. Lavanya, C. V. Meghana, S. Sanju, R. Shreyas, and S. Gagana, "A neural network model for wildfire scale prediction using meteorological factors," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 9, no. 6, pp. 2646–2650, Jun. 2021, doi: [10.22214/ijraset.2021.35258](https://doi.org/10.22214/ijraset.2021.35258).
- [75] S. Nosouhian, F. Nosouhian, and A. K. Khoshouei, "A review of recurrent neural network architecture for sequence learning: Comparison between LSTM and GRU," Preprints, MDPI, Basel, Switzerland, Tech. Rep., Jul. 2021, doi: [10.20944/preprints202107.0252.v1](https://doi.org/10.20944/preprints202107.0252.v1).
- [76] Y. Zhao, Y. Ban, and A. Nascetti, "Early detection of wildfires with GOES-R time-series and deep GRU network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 3765–3768, doi: [10.1109/IGARSS47720.2021.9554225](https://doi.org/10.1109/IGARSS47720.2021.9554225).
- [77] G. Mani and R. Volety, "A comparative analysis of LSTM and ARIMA for enhanced real-time air pollutant levels forecasting using sensor fusion with ground station data," *Cogent Eng.*, vol. 8, no. 1, Jan. 2021, Art. no. 1936886, doi: [10.1080/23311916.2021.1936886](https://doi.org/10.1080/23311916.2021.1936886).
- [78] X. Li, H. Gao, M. Zhang, S. Zhang, Z. Gao, J. Liu, S. Sun, T. Hu, and L. Sun, "Prediction of forest fire spread rate using UAV images and an LSTM model considering the interaction between fire and wind," *Remote Sens.*, vol. 13, no. 21, p. 4325, Oct. 2021, doi: [10.3390/rs13214325](https://doi.org/10.3390/rs13214325).
- [79] R. Ghosh and A. Kumar, "A hybrid deep learning model by combining convolutional neural network and recurrent neural network to detect forest fire," *Multimedia Tools Appl.*, vol. 81, no. 27, pp. 38643–38660, Nov. 2022, doi: [10.1007/s11042-022-13068-8](https://doi.org/10.1007/s11042-022-13068-8).
- [80] C. Herbert, J. F. Munoz-Martin, D. Llaveria, M. Pablos, and A. Camps, "Sea ice thickness estimation based on regression neural networks using L-band microwave radiometry data from the FSSCAT mission," *Remote Sens.*, vol. 13, no. 7, p. 1366, Apr. 2021, doi: [10.3390/rs13071366](https://doi.org/10.3390/rs13071366).
- [81] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. Adv. Neural Inf. Process. Syst.*, 2014, pp. 1–9. [Online]. Available: <http://www.github.com/goodfeli/adversarial>
- [82] O. G. Yalçin, "Generative adversarial network," in *Applied Neural Networks With TensorFlow 2*. Berkeley, CA, USA: Apress, 2021, pp. 259–284, doi: [10.1007/978-1-4842-6513-0_12](https://doi.org/10.1007/978-1-4842-6513-0_12).
- [83] S. P. H. Boroujeni and A. Razi, "IC-GAN: An improved conditional generative adversarial network for RGB-to-IR image translation with applications to forest fire monitoring," *Exp. Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 121962, doi: [10.1016/j.eswa.2023.121962](https://doi.org/10.1016/j.eswa.2023.121962).
- [84] M. H. Naveed, U. S. Hashmi, N. Tajved, N. Sultan, and A. Imran, "Assessing deep generative models on time series network data," *IEEE Access*, vol. 10, pp. 64601–64617, 2022, doi: [10.1109/ACCESS.2022.3177906](https://doi.org/10.1109/ACCESS.2022.3177906).
- [85] K. Yun, J. Bustos, and T. Lu, "Predicting rapid fire growth (Flashover) using conditional generative adversarial networks," *Electron. Imag.*, vol. 30, no. 9, pp. 127-1–127-4, Jan. 2018, doi: [10.2352/issn.2470-1173.2018.09.srv-127](https://doi.org/10.2352/issn.2470-1173.2018.09.srv-127).

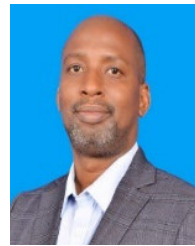
- [86] G. Wang, Y. Zhang, W. Xie, and Y. Qu, "Leveraging Google Earth engine and semi-supervised generative adversarial networks to assess initial burn severity in forest," *Can. J. Remote Sens.*, vol. 48, no. 3, pp. 411–424, May 2022, doi: [10.1080/07038992.2022.2054405](https://doi.org/10.1080/07038992.2022.2054405).
- [87] N. Q. Dung and H. Kim, "Generating high-resolution fire images with controllable attributes via generative adversarial networks," in *Proc. 22nd Int. Conf. Control, Autom. Syst. (ICCAS)*, Nov. 2022, pp. 348–353, doi: [10.23919/ICCAS55662.2022.10003687](https://doi.org/10.23919/ICCAS55662.2022.10003687).
- [88] A. E. Bakhshipour, A. Koochali, U. Dittmer, A. Haghighi, S. Ahmed, and A. Dengel, "A Bayesian generative adversarial networks (GAN) to generate synthetic time-series data, application in combined sewer flow prediction," in *Proc. 2nd Int. Join Conf. Water Distribution Syst. Anal. (WDSA) Comput. Control Water Ind. (CCWI)*, Jul. 2022, pp. 1–11, doi: [10.4995/wdsa-ccwi2022.2022.14699](https://doi.org/10.4995/wdsa-ccwi2022.2022.14699).
- [89] F.-J. Pérez-Porras, P. Triviño-Tarradas, C. Cima-Rodríguez, J.-E. Meroño-de-Larriba, A. García-Ferrer, and F.-J. Mesas-Carrascosa, "Machine learning methods and synthetic data generation to predict large wildfires," *Sensors*, vol. 21, no. 11, p. 3694, May 2021, doi: [10.3390/s21113694](https://doi.org/10.3390/s21113694).
- [90] A. Figueira and B. Vaz, "Survey on synthetic data generation, evaluation methods and GANs," *Mathematics*, vol. 10, no. 15, p. 2733, Aug. 2022, doi: [10.3390/math10152733](https://doi.org/10.3390/math10152733).
- [91] D. Saxena and J. Cao, "Generative adversarial networks (GANs)," *ACM Comput. Surv.*, vol. 54, no. 3, pp. 1–42, Apr. 2022, doi: [10.1145/3446374](https://doi.org/10.1145/3446374).
- [92] H. Chen, "Challenges and corresponding solutions of generative adversarial networks (GANs): A survey study," *J. Phys., Conf.*, vol. 1827, no. 1, Mar. 2021, Art. no. 012066, doi: [10.1088/1742-6596/1827/1/012066](https://doi.org/10.1088/1742-6596/1827/1/012066).
- [93] A. Shmuel and E. Heifetz, "Global wildfire susceptibility mapping based on machine learning models," *Forests*, vol. 13, no. 7, p. 1050, Jul. 2022, doi: [10.3390/f13071050](https://doi.org/10.3390/f13071050).
- [94] K. Gao, Z. Feng, and S. Wang, "Using multilayer perceptron to predict forest fires in Jiangxi Province, southeast China," *Discrete Dyn. Nature Soc.*, vol. 2022, no. 1, Jan. 2022, Art. no. 6930812, doi: [10.1155/2022/6930812](https://doi.org/10.1155/2022/6930812).
- [95] M. L. Pai, K. S. Varsha, and R. Arya, "Application of artificial neural networks and genetic algorithm for the prediction of forest fire danger in Kerala," in *Advances in Intelligent Systems and Computing*, vol. 941. Cham, Switzerland: Springer, 2020, pp. 935–942, doi: [10.1007/978-3-030-16660-1_91](https://doi.org/10.1007/978-3-030-16660-1_91).
- [96] P. Jain, S. C. P. Coogan, S. G. Subramanian, M. Crowley, S. Taylor, and M. D. Flannigan, "A review of machine learning applications in wildfire science and management," *Environ. Rev.*, vol. 28, no. 4, pp. 478–505, Dec. 2020, doi: [10.1139/er-2020-0019](https://doi.org/10.1139/er-2020-0019).
- [97] V. Khryashchev and R. Larionov, "Wildfire segmentation on satellite images using deep learning," in *Proc. Moscow Workshop Electron. Netw. Technol. (MWENT)*, Moscow, Russia, Mar. 2020, pp. 1–5, doi: [10.1109/MWENT47943.2020.9067475](https://doi.org/10.1109/MWENT47943.2020.9067475).
- [98] D. Rashkovetsky, F. Mauracher, M. Langer, and M. Schmitt, "Wildfire detection from multisensor satellite imagery using deep semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 7001–7016, 2021, doi: [10.1109/JSTARS.2021.3093625](https://doi.org/10.1109/JSTARS.2021.3093625).
- [99] L. Knopp, M. Wieland, M. Rättich, and S. Martinis, "A deep learning approach for burned area segmentation with Sentinel-2 data," *Remote Sens.*, vol. 12, no. 15, p. 2422, Jul. 2020, doi: [10.3390/rs12152422](https://doi.org/10.3390/rs12152422).
- [100] F. Huot, R. Lily Hu, M. Ihme, Q. Wang, J. Burge, T. Lu, J. Hickey, Y.-F. Chen, and J. Anderson, "Deep learning models for predicting wildfires from historical remote-sensing data," 2020, *arXiv:2010.07445*.
- [101] F. Khennou and M. A. Akhloufi, "Improving wildland fire spread prediction using deep U-Nets," *Sci. Remote Sens.*, vol. 8, Dec. 2023, Art. no. 100101, doi: [10.1016/j.srs.2023.100101](https://doi.org/10.1016/j.srs.2023.100101).
- [102] S. Sirajudeen, "An overview of you only look once: Unified, real-time object detection," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 8, no. 6, pp. 607–609, Jun. 2020, doi: [10.22214/ijraset.2020.6098](https://doi.org/10.22214/ijraset.2020.6098).
- [103] J. Lee and K.-I. Hwang, "YOLO with adaptive frame control for real-time object detection applications," *Multimedia Tools Appl.*, vol. 81, no. 25, pp. 36375–36396, Oct. 2022, doi: [10.1007/s11042-021-11480-0](https://doi.org/10.1007/s11042-021-11480-0).
- [104] L. Zhao, L. Zhi, C. Zhao, and W. Zheng, "Fire-YOLO: A small target object detection method for fire inspection," *Sustainability*, vol. 14, no. 9, p. 4930, Apr. 2022, doi: [10.3390/su14094930](https://doi.org/10.3390/su14094930).
- [105] E. Ajitha, V. Priya, and D. R. I. Dhashini, "Visualization of real time forest fire detection," in *Proc. 3rd Int. Conf. Intell. Techn. Control, Optim. Signal Process. (INCOS)*, Mar. 2024, pp. 1–5, doi: [10.1109/incos59338.2024.10527658](https://doi.org/10.1109/incos59338.2024.10527658).
- [106] H. Zheng, S. Dembélé, Y. Wu, Y. Liu, H. Chen, and Q. Zhang, "A lightweight algorithm capable of accurately identifying forest fires from UAV remote sensing imagery," *Frontiers Forests Global Change*, vol. 6, Feb. 2023, Art. no. 1134942, doi: [10.3389/ffgc.2023.1134942](https://doi.org/10.3389/ffgc.2023.1134942).
- [107] A. Safonova, G. Ghazaryan, S. Stiller, M. Main-Knorn, C. Nendel, and M. Ryo, "Ten deep learning techniques to address small data problems with remote sensing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 125, Dec. 2023, Art. no. 103569, doi: [10.1016/j.jag.2023.103569](https://doi.org/10.1016/j.jag.2023.103569).
- [108] A. Saleh, M. A. Zulkifley, H. H. Harun, F. Gaudreault, I. Davison, and M. Spraggon, "Forest fire surveillance systems: A review of deep learning methods," *Heliyon*, vol. 10, no. 1, Jan. 2024, Art. no. e23127, doi: [10.1016/j.heliyon.2023.e23127](https://doi.org/10.1016/j.heliyon.2023.e23127).
- [109] S. T. Seydi, V. Saeidi, B. Kalantar, N. Ueda, and A. A. Halin, "Fire-net: A deep learning framework for active forest fire detection," *J. Sensors*, vol. 2022, pp. 1–14, Feb. 2022, doi: [10.1155/2022/8044390](https://doi.org/10.1155/2022/8044390).
- [110] V. Neethidevan and S. Anand, "Implementing and evaluating the performance of various machine learning algorithms with different datasets," *Int. J. Health Sci.*, vol. 6, no. 1, pp. 4684–4694, Apr. 2022, doi: [10.53730/ijhs.v6ns1.5890](https://doi.org/10.53730/ijhs.v6ns1.5890).
- [111] N. Omar, A. Al-zebari, and A. Sengur, "Deep learning approach to predict forest fires using meteorological measurements," in *Proc. 2nd Int. Informat. Softw. Eng. Conf. (IISEC)*, Dec. 2021, pp. 1–4, doi: [10.1109/IISEC54230.2021.9672446](https://doi.org/10.1109/IISEC54230.2021.9672446).
- [112] Z. Chen, C. Zhang, W. Li, L. Gao, L. Liu, L. Fang, and C. Zhang, "Fire danger forecasting using machine learning-based models and meteorological observation: A case study in Northeastern China," *Multimedia Tools Appl.*, vol. 83, no. 22, pp. 61861–61881, Jun. 2023, doi: [10.1007/s11042-023-15881-1](https://doi.org/10.1007/s11042-023-15881-1).
- [113] Y. Gupta, N. Goyal, V. J. Varghese, and P. Goyal, "Utilizing MODIS fire mask for predicting forest fires using Landsat-9/8 and meteorological data," in *Proc. IEEE 10th Int. Conf. Data Sci. Adv. Anal.*, 2023, pp. 1–10, doi: [10.1109/DSAA60987.2023.10302582](https://doi.org/10.1109/DSAA60987.2023.10302582).
- [114] D. Radke, A. Hessler, and D. Ellsworth. (2019). *FireCast: Leveraging Deep Learning To Predict Wildfire Spread*. [Online]. Available: <https://glovis.usgs.gov/app?fullscreen=0>
- [115] J. Deng, W. Wang, G. Gu, Z. Chen, J. Liu, G. Xie, S. Weng, L. Ding, and C. Li, "Wildfire susceptibility prediction using a multisource and spatiotemporal cooperative approach," *Earth Sci. Informat.*, vol. 16, no. 4, pp. 3511–3529, Dec. 2023, doi: [10.1007/s12145-023-01104-6](https://doi.org/10.1007/s12145-023-01104-6).
- [116] C. C. Joshi, J. S. S. K. Payyavula, S. Patel, and Y. M. Alginahi, "ML-based wildfire prediction and detection," in *Proc. IEEE 3rd Int. Conf. Comput. Mach. Intell.*, Apr. 2024, pp. 1–5, doi: [10.1109/ICMI60790.2024.10585687](https://doi.org/10.1109/ICMI60790.2024.10585687).
- [117] A. Mutakabbir, C.-H. Lung, S. A. Ajila, M. Zaman, K. Naik, R. Purcell, and S. Sampalli, "Spatio-temporal agnostic deep learning modeling of forest fire prediction using weather data," in *Proc. IEEE 47th Annu. Comput., Softw., Appl. Conf. (COMPSAC)*, Jun. 2023, pp. 346–351, doi: [10.1109/compsac57700.2023.00054](https://doi.org/10.1109/compsac57700.2023.00054).
- [118] S. Bhatt and U. Chouhan, "An enhanced method for predicting and analysing forest fires using an attention-based CNN model," *J. Forestry Res.*, vol. 35, no. 1, p. 67, Dec. 2024, doi: [10.1007/s11676-024-01717-7](https://doi.org/10.1007/s11676-024-01717-7).
- [119] X. You, Z. Zheng, K. Yang, L. Yu, J. Liu, J. Chen, X. Lu, and S. Guo, "A PSO-CNN-based deep learning model for predicting forest fire risk on a national scale," *Forests*, vol. 15, no. 1, p. 86, Dec. 2023, doi: [10.3390/f15010086](https://doi.org/10.3390/f15010086).
- [120] S. Chilveri, T. Jain, and P. H. Mhaske, "Satellite image processing using radiometric resolution & spectral resolution," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 11, no. 3, pp. 112–120, Mar. 2023, doi: [10.22214/ijraset.2023.49300](https://doi.org/10.22214/ijraset.2023.49300).
- [121] C. J. Seaman, W. Line, R. Ziel, J. Jenkins, C. Dierking, and G. Hanson, "Multispectral satellite imagery products for fire weather applications," *J. Atmos. Ocean. Technol.*, vol. 40, no. 6, pp. 719–738, Jun. 2023, doi: [10.1175/jtech-d-22-0107.1](https://doi.org/10.1175/jtech-d-22-0107.1).
- [122] A. Pepe, D. Stroppiana, F. Calo, P. Imperatore, L. Boschetti, C. Bignami, P. A. Brivio, and R. Lanari, "Exploitation of Copernicus sentinels data for sensing fire-disturbed vegetated areas," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 7589–7592, doi: [10.1109/IGARSS.2018.8518272](https://doi.org/10.1109/IGARSS.2018.8518272).
- [123] Y. Ji, D. Wang, Q. Li, T. Liu, and Y. Bai, "Global wildfire danger predictions based on deep learning taking into account static and dynamic variables," *Forests*, vol. 15, no. 1, p. 216, Jan. 2024, doi: [10.3390/f15010216](https://doi.org/10.3390/f15010216).
- [124] United States Geological Survey (USGS). *Application—GloVis*. Accessed: Aug. 3, 2024. [Online]. Available: <https://glovis.usgs.gov/app?fullscreen=0>
- [125] M. J. Wooster, G. J. Roberts, L. Giglio, D. P. Roy, P. H. Freeborn, L. Boschetti, C. Justice, C. Ichoku, W. Schroeder, D. Davies, and A. M. Smith, "Satellite remote sensing of active fires: History and current status, applications and future requirements," *Remote Sens. Environ.*, vol. 267, Dec. 2021, Art. no. 112694, doi: [10.1016/j.rse.2021.112694](https://doi.org/10.1016/j.rse.2021.112694).

- [126] A. Dakir, B. F. Zahra, and A. B. Omar, "Optical satellite images services for precision agricultural use: A review," *Adv. Sci., Technol. Eng. Syst. J.*, vol. 6, no. 3, pp. 326–331, Jun. 2021, doi: [10.25046/aj060337](#).
- [127] NASAOffice. MODISatellite. *MODIS Web*. Accessed: Aug. 3, 2024. [Online]. Available: <https://modis.gsfc.nasa.gov/about/>
- [128] D. Radoč aj, M. Jurišić, and M. Gašparović, "A wildfire growth prediction and evaluation approach using Landsat and MODIS data," *J. Environ. Manage.*, vol. 304, Feb. 2022, Art. no. 114351, doi: [10.1016/j.jenvman.2021.114351](#).
- [129] A. Comerón, C. Muñoz-Porcar, F. Rocadenbosch, A. Rodríguez-Gómez, and M. Sicard, "Current research in lidar technology used for the remote sensing of atmospheric aerosols," *Sensors*, vol. 17, no. 6, p. 1450, Jun. 2017, doi: [10.3390/s17061450](#).
- [130] D. Monaco. *GitHub—Dbdm/Rescue: Pytorch Implementation of the Double-Step Framework to Perform Wildfires Severity Estimation From Sentinel-2 Satellite Images*. Accessed: Aug. 3, 2024. [Online]. Available: <https://github.com/dbdm/rescue>
- [131] A. Farasin, L. Colomba, and P. Garza, "Double-step U-net: A deep learning-based approach for the estimation of wildfire damage severity through Sentinel-2 satellite data," *Appl. Sci.*, vol. 10, no. 12, p. 4332, Jun. 2020, doi: [10.3390/app10124332](#).
- [132] E. P. Yankovich, K. S. Yankovich, and N. V. Baranovskiy, "Forest mapping using classification of Sentinel-2A imagery for forest fire danger prediction: A case study," *Int. J. Eng. Appl. (IREA)*, vol. 9, no. 3, pp. 148–161, May 2021, doi: [10.15866/irea.v9i3.20209](#).
- [133] S. Arjasakusuma, S. S. Kusuma, Y. Vetrata, I. Prasasti, and R. Arief, "Monthly burned-area mapping using multi-sensor integration of Sentinel-1 and Sentinel-2 and machine learning: Case study of 2019's fire events in South Sumatra Province, Indonesia," *Remote Sens. Applications: Soc. Environ.*, vol. 27, Aug. 2022, Art. no. 100790, doi: [10.1016/j.rsase.2022.100790](#).
- [134] Japan Aerospace Exploration Agency. Himawari-8 Dataset. *JAXA Himawari Monitor | User's Guide*. Accessed: Aug. 3, 2024. [Online]. Available: <https://www.eorc.jaxa.jp/tree/userguide.html>
- [135] K. Bessho, H. Owada, K. Okamoto, and T. Fujita, "Himawari-8/9 follow-on satellite program and impacts of potential usage of hyperspectral IR sounder," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 1507–1510, doi: [10.1109/IGARSS47720.2021.9553888](#).
- [136] United States Geological Survey (USGS). Geospatial Multi-Agency Coordination (GeoMAC). *Wildland Fire Map, GeoMAC | U.S. Geological Survey*. Accessed: Aug. 3, 2024. [Online]. Available: <https://www.usgs.gov/media/images/wildland-fire-map-geomac>
- [137] NOAA Office of Satellite and Product Operations. GOES Satellite. *GOES | OSPO*. Accessed: Aug. 3, 2024. [Online]. Available: <https://www.ospo.noaa.gov/operations/goes/transition.html>
- [138] T. N. Toan, T. P. Cong, Q. N. Viet Hung, and J. Jo, "A deep learning approach for early wildfire detection from hyperspectral satellite images," in *Proc. 7th Int. Conf. Robot Intell. Technol. Appl. (RITA)*, 2019, pp. 38–45.
- [139] B. T. Pham, A. Jaafari, M. Avand, N. Al-Ansari, T. D. Du, H. P. H. Yen, T. V. Phong, D. H. Nguyen, H. V. Le, D. Mafi-Gholami, I. Prakash, H. T. Thuy, and T. T. Tuyen, "Performance evaluation of machine learning methods for forest fire modeling and prediction," *Symmetry*, vol. 12, no. 6, p. 1022, Jun. 2020, doi: [10.3390/sym12061022](#).
- [140] Y. Kountouris, "Human activity, daylight saving time and wildfire occurrence," *Sci. Total Environ.*, vol. 727, Jul. 2020, Art. no. 138044, doi: [10.1016/j.scitotenv.2020.138044](#).
- [141] N. de Magalhães Neto and H. Evangelista, "Human activity behind the unprecedented 2020 wildfire in Brazilian wetlands (Pantanal)," *Frontiers Environ. Sci.*, vol. 10, Jun. 2022, Art. no. 888578, doi: [10.3389/fenvs.2022.888578](#).
- [142] N. Phelps and D. G. Woolford, "Comparing calibrated statistical and machine learning methods for wildland fire occurrence prediction: A case study of human-caused fires in Lac La Biche, Alberta, Canada," *Int. J. Wildland Fire*, vol. 30, no. 11, pp. 850–870, Sep. 2021, doi: [10.1071/wf20139](#).
- [143] D. Swetha, P. Deepa, P. Deepthy, N. M. Raval, and M. Nandini, "Forest fire prediction using IoT and deep learning," *Int. J. Innov. Res. Technol.*, vol. 9, no. 11, pp. 313–317, Apr. 2023.
- [144] U. Dampage, L. Bandaranayake, R. Wanasinghe, K. Kottahachchi, and B. Jayasanka, "Forest fire detection system using wireless sensor networks and machine learning," *Sci. Rep.*, vol. 12, no. 1, p. 46, Jan. 2022, doi: [10.1038/s41598-021-03882-9](#).
- [145] J. Ananthi, N. Sengottaiyan, S. Anbukaruppusamy, K. Upreti, and A. K. Dubey, "Forest fire prediction using IoT and deep learning," *Int. J. Adv. Technol. Eng. Explor.*, vol. 9, no. 87, pp. 246–256, Feb. 2022, doi: [10.19101/IJATEE.2021.87464](#).
- [146] J. D. L. P. Irakomeye, A. Nshimiye, K. Ng'ondya, E. J. Sumari, D. J. Leo, and D. A. Sam, "Development of IoT based security system for fire monitoring and prevention in Tanzanian industries," *Int. J. Adv. Sci. Res. Eng.*, vol. 7, no. 12, pp. 38–44, 2022, doi: [10.31695/ijasre.2021.34120](#).
- [147] R. Ghali, M. Jmal, W. Soudiene Mseddi, and R. Attia, "Recent advances in fire detection and monitoring systems: A review," in *Proc. 8th Int. Conf. Sci. Electron., Technol. Inf. Telecommun.*, 2020, pp. 332–340, doi: [10.1007/978-3-030-21005-2_32](#).
- [148] Scopus. *Scopus | Abstract and Citation Database*. Accessed: Mar. 27, 2024. [Online]. Available: <https://www.elsevier.com/products/scopus>
- [149] B. Kitchenham, O. Pearl Brereton, D. Budgen, M. Turner, J. Bailey, and S. Linkman, "Systematic literature reviews in software engineering—A systematic literature review," *Inf. Softw. Technol.*, vol. 51, no. 1, pp. 7–15, Jan. 2009, doi: [10.1016/j.infsof.2008.09.009](#).
- [150] I. Prapas, S. Kondylatos, I. Papoutsis, G. Camps-Valls, M. Ronco, M.-Á. Fernández-Torres, M. P. Guillem, and N. Carvalhais, "Deep learning methods for daily wildfire danger forecasting," 2021, *arXiv:2111.02736*.
- [151] M. Lindemann, K. DeMichele, M. H. Kapourchali, M. H. Kapourchali, C. Waigl, E. Trochim, and L. Zhao, "Igniting precision: Amplifying wildfire prediction in diverse regions via teacher-student model fusion," in *Proc. Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2023, pp. 528–535, doi: [10.1109/icmla58977.2023.00079](#).
- [152] P. Kaur, K. Naik, R. Purcell, S. Sampalli, C.-H. Lung, M. Zaman, and A. Mutakabbir, "A data integration framework with multi-source big data for enhanced forest fire prediction," in *Proc. IEEE Int. Conf. Big Data (BigData)*, Dec. 2023, pp. 344–351, doi: [10.1109/big-data59044.2023.10386816](#).
- [153] J. R. Bergado, C. Persello, K. Reinke, and A. Stein, "Predicting wildfire burns from big geodata using deep learning," *Saf. Sci.*, vol. 140, Aug. 2021, Art. no. 105276, doi: [10.1016/j.ssci.2021.105276](#).
- [154] K. Pacifici, B. J. Reich, D. A. W. Miller, and B. S. Pease, "Resolving misaligned spatial data with integrated species distribution models," *Ecology*, vol. 100, no. 6, Jun. 2019, Art. no. e02709, doi: [10.1002/ecy.2709](#).
- [155] Y. Zou, M. Sadeghi, Y. Liu, A. Puchko, S. Le, Y. Chen, N. Andela, and P. Gentine, "Attention-based wildland fire spread modeling using fire-tracking satellite observations," *Fire*, vol. 6, no. 8, p. 289, Jul. 2023, doi: [10.3390/fire6080289](#).
- [156] A. Larsen, I. Hanigan, B. J. Reich, Y. Qin, M. Cope, G. Morgan, and A. G. Rappold, "A deep learning approach to identify smoke plumes in satellite imagery in near-real time for health risk communication," *J. Exposure Sci. Environ. Epidemiology*, vol. 31, no. 1, pp. 170–176, Jan. 2021, doi: [10.1038/s41370-020-0246-y](#).
- [157] T. H. Sanford, L. Zhang, S. A. Harmon, J. Sackett, D. Yang, H. Roth, Z. Xu, D. Kesani, S. Mehralivand, R. H. Baroni, T. Barrett, R. Girometti, A. Oto, A. S. Purysko, S. Xu, P. A. Pinto, D. Xu, B. J. Wood, P. L. Choyke, and B. Turkbey, "Data augmentation and transfer learning to improve generalizability of an automated prostate segmentation model," *Amer. J. Roentgenol.*, vol. 215, no. 6, pp. 1403–1410, Dec. 2020, doi: [10.2214/ajr.19.22347](#).
- [158] J. Su, X. Yu, X. Wang, Z. Wang, and G. Chao, "Enhanced transfer learning with data augmentation," *Eng. Appl. Artif. Intell.*, vol. 129, Mar. 2024, Art. no. 107602, doi: [10.1016/j.engappai.2023.107602](#).
- [159] M. Gusenbauer and N. R. Haddaway, "Which academic search systems are suitable for systematic reviews or meta-analyses? Evaluating retrieval qualities of Google scholar, PubMed, and 26 other resources," *Res. Synth. Methods*, vol. 11, no. 2, pp. 181–217, Mar. 2020, doi: [10.1002/jrsm.1378](#).
- [160] D. Mane, A. Magar, O. Khode, S. Koli, K. Bhat, and P. Korade, "Unlocking machine learning model decisions: A comparative analysis of LIME and SHAP for enhanced interpretability," *J. Electr. Syst.*, vol. 20, no. 2, pp. 598–613, Apr. 2024, doi: [10.52783/jes.1480](#).
- [161] R. Mbuvha, Y. Yaakoubi, J. Bagiliko, S. H. Potes, A. Nammouchi, and S. S. Amrouche, "Leveraging AI for climate resilience in africa: Challenges, opportunities, and the need for collaboration," *SSRN Electron. J.*, May 2024, doi: [10.2139/ssrn.4815919](#).
- [162] F. H. Muttaqien, T. Wirahman, A. Shabrina, and A. L. Latifah, "Interactive simulation tool for spatiotemporal data of climate-related fires over Borneo," in *Proc. 11th Int. Conf. Inf. Commun. Technol. (ICOICT)*, Aug. 2023, pp. 367–372, doi: [10.1109/icoict58202.2023.10262511](#).

- [163] M. Yandouzi, M. Grari, I. Idrissi, O. Moussaoui, M. Azizi, K. Ghoumid, and A. K. Elmiad, "Review on forest fires detection and prediction using deep learning and drones," *J. Theor. Appl. Inf. Technol.*, vol. 100, no. 19, pp. 5450–5457, Oct. 2022, [Online]. Available: www.jatit.org
- [164] N. A. Khairani and E. Sutoyo, "Application of K-means clustering algorithm for determination of fire-prone areas utilizing hotspots in West Kalimantan Province," *Int. J. Adv. Data Inf. Syst.*, vol. 1, no. 1, pp. 9–16, Apr. 2020, doi: [10.25008/ijadis.v1i1.13](https://doi.org/10.25008/ijadis.v1i1.13).
- [165] J. Sharma, P.-A. Andersen, O.-C. Granmo, and M. Goodwin, "Deep Q-learning with Q-matrix transfer learning for novel fire evacuation environment," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 51, no. 12, pp. 7363–7381, Dec. 2021, doi: [10.1109/TSMC.2020.2967936](https://doi.org/10.1109/TSMC.2020.2967936).
- [166] L. Lv, T. Chen, J. Dou, and A. Plaza, "A hybrid ensemble-based deep-learning framework for landslide susceptibility mapping," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 108, Apr. 2022, Art. no. 102713, doi: [10.1016/j.jag.2022.102713](https://doi.org/10.1016/j.jag.2022.102713).



CESILIA MAMBILE (Member, IEEE) received the master's degree in information and communication sciences and engineering from The Nelson Mandela African Institution of Science and Technology (NM-AIST), where she is currently pursuing the Ph.D. degree in information and communication sciences and engineering. She is a Computer Scientist and an Assistant Lecturer with the College of Informatics and Virtual Education (CIVE), The University of Dodoma. Her research interests include addressing climate change by using artificial intelligence and machine learning. She is also developing a deep learning model using satellite images to predict forest fire occurrence linked with seasonality in Mount Kilimanjaro, Tanzania.



SHUBI KAIJAGE (Senior Member, IEEE) received the Ph.D. degree in electronics and information engineering from the University of Ryukyus, Japan, in 2011. For over five years, since March 2017, he has been the Dean of the School of Computational and Communication Science and Engineering (CoCSE), The Nelson Mandela African Institution of Science and Technology (NM-AIST), Tanzania. He is currently an Associate Professor of electronics engineering and information systems. His research interests include optical fibers and photonic crystal fibers (PCFs), fiber optics communication, fiber-to-the-home (FTTH), terahertz wave technology, radio frequency identification (RFID), the Internet of Things (IoT), and wireless sensor networks. Additionally, he is interested in exploring the application of artificial intelligence (AI), electronics, and information systems in the areas of e-health management, conservation, and protection of natural resources, climate change, livestock and sustainable agriculture, water, food security, and sustainability. He is a Senior Member of African Academy of Science (AAS). He has published more than 80 scholarly articles and earned numerous scholarly awards. He has been serving on different committees, such as the General Chair for the 2021 IEEE AFRICON International Conference, School/Faculty Boards, Senate, and other institutional and external committees. Shubi Google Scholar Shubi's Web of Science



JUDITH LEO (Member, IEEE) received the bachelor's degree in engineering in electronics and ICT, majoring in engineering from Visvesvaraya Technological University, Karnataka, India, in 2009, the M.Sc. degree in information and communication science and engineering, majoring in information technology system development and management, mobile apps, and geographical information systems and the Ph.D. degree in information and communication science and engineering, majoring in machine learning and the Internet of Things from The Nelson Mandela African Institution of Science and Technology (NM-AIST), Arusha, Tanzania, in 2017 and 2020, respectively.

She is currently a Senior Lecturer with the School of Computation and Communication Science and Engineering, NM-AIST. She has experience working with multidisciplinary research actions that integrate diverse areas in information and communication technology; gender inclusion and policies; leadership in STEMI; and environment, health, social, and business areas. Her research interests include data science and data-driven solutions, the use of artificial intelligence (AI), machine learning (ML), mobile apps, the Internet of Things (IoT) to solve societal and industrial challenges, the area of optical fiber installation, bandwidth, energy optimization, advanced network configuration, system development, and modeling. She won several grants, awards, and scholarships.

...